

EASY WAY TO FIND UNIT PLACE OF m^n WHERE m, n ARE NATURAL NUMBERS

Let m^n is the given number, where m, n are natural numbers. Here we can find the unit place of m^n by a simple method.

Process: -

Unit place of $m^n = \text{Unit place of (Unit place of } m)^n = \text{Unit place of } (a^n)$

Where 'a' is unit place of m and $a \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

1. If $a = 0$, then

Unit place of $(a^n) = 0$

2. If $a = 1$, then

Unit place of $(a^n) = 1$

3. If $a = 2$, then

$$\text{Unit place of } (a^n) = \begin{cases} 2 & \text{if } n = 4k + 1 \\ 4 & \text{if } n = 4k + 2 \\ 8 & \text{if } n = 4k + 3 \\ 6 & \text{if } n = 4k \end{cases}$$

Here k is a natural number.

4. If $a = 3$, then

$$\text{Unit place of } (a^n) = \begin{cases} 3 & \text{if } n = 4k + 1 \\ 9 & \text{if } n = 4k + 2 \\ 7 & \text{if } n = 4k + 3 \\ 1 & \text{if } n = 4k \end{cases}$$

Here k is a natural number.

5. If $a = 4$, then

$$\text{Unit place of } (a^n) = \begin{cases} 4 & \text{if } n = 2k + 1 \\ 6 & \text{if } n = 2k \end{cases}$$

Here k is a natural number.

6. If $a = 5$, then

Unit place of $(a^n) = 5$

7. If $a = 6$, then

Unit place of $(a^n) = 6$

8. If $a = 7$, then

$$\text{Unit place of } (a^n) = \begin{cases} 7 & \text{if } n = 4k + 1 \\ 9 & \text{if } n = 4k + 2 \\ 3 & \text{if } n = 4k + 3 \\ 1 & \text{if } n = 4k \end{cases}$$

Here k is a natural number.

9. If $a = 8$, then

$$\text{Unit place of } (a^n) = \begin{cases} 8 & \text{if } n = 4k + 1 \\ 4 & \text{if } n = 4k + 2 \\ 2 & \text{if } n = 4k + 3 \\ 6 & \text{if } n = 4k \end{cases}$$

Here k is a natural number.

10. If $a = 9$, then

$$\text{Unit place of } (a^n) = \begin{cases} 9 & \text{if } n = 2k + 1 \\ 1 & \text{if } n = 2k \end{cases}$$

Here k is a natural number.

Example 1: - Find the unit place in the expansion of 3897654^{876548} .

$$\begin{aligned} \text{Ans: - Unit place of } 3897654^{876548} &= \text{Unit place of (Unit place of } (3897654) \text{)}^{876548} \\ &= \text{Unit place of } (4)^{876548} \\ &= \text{Unit place of } (4)^{2 \times 438274} \\ &= \text{Unit place of } (4)^{2 \times k} \\ &= 6 \end{aligned}$$

Example 2: - Find the unit place in the expansion of 653768957^{76531} .

$$\begin{aligned} \text{Ans: - Unit place of } (653768957)^{76531} &= \text{Unit place of (Unit place of } (653768957) \text{)}^{76531} \\ &= \text{Unit place of } (7)^{76531} \\ &= \text{Unit place of } (7)^{4 \times 19132 + 3} \\ &= \text{Unit place of } (7)^{4 \times k + 3} \\ &= 3 \end{aligned}$$

Example 3: - Find the unit place in the expansion of $689^{745} + 745^{689}$

$$\begin{aligned} \text{Ans: - Unit place of } 689^{745} &= \text{Unit place of (Unit place of } (689) \text{)}^{745} \\ &= \text{Unit place of } (9)^{745} \\ &= \text{Unit place of } (9)^{2 \times 372 + 1} \\ &= \text{Unit place of } (9)^{2 \times k + 1} \\ &= 9 \end{aligned}$$

$$\begin{aligned} \text{Unit place of } 745^{689} &= \text{Unit place of (Unit place of } (745) \text{)}^{689} \\ &= \text{Unit place of } (5)^{689} \\ &= 5 \end{aligned}$$

$$\begin{aligned} \text{Unit place of } 689^{745} + 745^{689} &= \text{Unit place of } 689^{745} + \text{Unit place of } 745^{689} \\ &= \text{Unit Place of } (9+5) \\ &= \text{Unit Place of } 14 \\ &= 4 \text{ (Ans).} \end{aligned}$$

Hence in this way we can calculate unit place of any natural number to the power any natural number.

Exercise:

1. Find unit place of $(243768953)^{145533}$.
2. Find unit place of $(12300457)^{345627}$.
3. Find unit place of $(523432959)^{983456}$.
4. Find unit place of $(223432)^{733406}$.
5. Find unit place of $(329595234)^{456983}$.
6. Find unit place of $(7689511)^{334455}$.
7. Find unit place of $(76895)^{100034}$.
8. Find unit place of $(1024376)^{99806}$.
9. Find unit place of $(872790)^{4550}$.
10. Find unit place of $(663768)^{663768}$.
11. Find unit place of $(243)^{533} - (65)^{342}$.
12. Find unit place of $(243)^{533} + (65)^{342}$.
13. Find unit place of $(243)^{533} \times (65)^{342}$.

CLOCKS

In this topic basic idea is what is angle between minute hand and hour hand at a particular time. In every clock there are three hands namely Hour hand, Minute hand and Second hand.

1. Hour hand describes an angle 360° in 12 hours. So in 1 hour, Hour hand describe an angle of 30° .
2. Minute hand describes an angle 360° in 60 minutes. So in 1 minute, Minute hand describe an angle of 6° .
3. Second hand describes an angle 360° in 60 seconds. So in 1 second, Second hand describe an angle of 6° .
4. At 12 O' Clock Hour hand and Minute hand coincides. Angle between them is 0° .
5. At 6 O' Clock Hour hand and Minute hand are opposite to each other. Angle between them is 180° .
6. At 3 O' Clock and 9 O' Clock Hour hand and Minute hands are perpendicular to each other. Angle between them is 90° .
7. Clock is divided into 60 equal divisions called as minute divisions.
8. In one hour the minute hand moves 60 divisions and hour hand moves 5 divisions. Hence Minute hand gains 55 minutes over the hour hand in 60 minutes or in one hour.

Q. At what time after 5 O' Clock both Hour and Minute hand coincide with each other?

Q. At what time between 5 O' Clock and 6 O' Clock both Hour and Minute hand coincide with each other?

Ans: At 5 O' Clock the minute hand is at 12 and hour hand is at 5.

So, the minute hand is 25 minutes behind the hour hand.

Both Hour and Minute hand coincide with each other if minute hand gains 25 minutes over the hour hand.

As we know Minute hand gains 55 minutes over the hour hand in 60 minutes.

So, Minute hand gains 1 minute over the hour hand in $\frac{60}{55}$ minutes.

Minute hand gains 25 minutes over the hour hand in $\frac{60}{55} \times 25 = \frac{300}{11} = 27\frac{3}{11}$ minutes.

Hence both the hands coincide at 5 hour $27\frac{3}{11}$ minutes.

Q. At what time after 5 O' Clock both Hour and Minute hand are perpendicular to each other?

Q. At what time between 5 O' Clock and 6 O' Clock both Hour and Minute hand are perpendicular to each other?

Ans: At 5 O' Clock the minute hand is at 12 and hour hand is at 5.

So, the minute hand is 25 minutes or 150° behind the hour hand.

Both Hour and Minute hand coincide are perpendicular to each other if minute hand gains $150^\circ + 90^\circ = 240^\circ$ or $150^\circ - 90^\circ = 60^\circ$ i.e it gains 40 minutes or 10 minutes over the hour hand.

(i) As we know Minute hand gains 55 minutes over the hour hand in 60 minutes.

So, Minute hand gains 1 minute over the hour hand in $\frac{60}{55}$ minutes.

Minute hand gains 40 minutes over the hour hand in $\frac{60}{55} \times 40 = \frac{480}{11} = 43\frac{7}{11}$ minutes.

(ii) As we know Minute hand gains 55 minutes over the hour hand in 60 minutes.

So, Minute hand gains 1 minute over the hour hand in $\frac{60}{55}$ minutes.

Minute hand gains 10 minutes over the hour hand in $\frac{60}{55} \times 10 = \frac{120}{11} = 10\frac{10}{11}$ minutes.

Hence both the hands are perpendicular to each other at 5 hour $10\frac{10}{11}$ minutes and 5 hour $43\frac{7}{11}$ minutes

Q. At what time after 4 O' Clock both Hour and Minute hand are opposite to each other?

Q. At what time between 4 O' Clock and 5 O' Clock both Hour and Minute hand are opposite to each other?

Ans: At 4 O' Clock the minute hand is at 12 and hour hand is at 4.

So, the minute hand is 20 minutes or 120° behind the hour hand.

Both Hour and Minute hand will be opposite to each other if minute hand gains $120^\circ + 180^\circ = 300^\circ$ i.e. 50 minutes over the hour hand.

As we know Minute hand gains 55 minutes over the hour hand in 60 minutes.

So, Minute hand gains 1 minute over the hour hand in $\frac{60}{55}$ minutes.

Minute hand gains 50 minutes over the hour hand in $\frac{60}{55} \times 50 = \frac{600}{11} = 54\frac{6}{11}$ minutes.

Hence both the hands are opposite to each other at 4 hour $54\frac{6}{11}$ minutes.

Q. A person left home between 4 PM and 5 PM and returned home between 5 PM and 6 PM and found that his Hour and Minute hand are changed their places. When did he left the home?

Ans: Hour hand is in x minute division i.e. $20 < x < 25$ and when he returned back then hour hand is at y minute division i.e. $25 < y < 30$.

As minute hand moves 12 times faster than hour hand. Therefore considering the position of the minute hand at the time of going out and returned back we have:

$$y = 12(x - 20) \text{ and } x = 12(y - 25)$$

$$\Rightarrow 12x - y = 240 \text{ and } x - 12y = -300$$

$$\text{After solving } x = \frac{3180}{143} \text{ and } y = \frac{3840}{143} = 26\frac{122}{143}.$$

Hence he left the home at 4 hour $26\frac{122}{143}$ minutes.

Exercise:

- 1. At what time after 8 O' Clock both Hour and Minute hand coincide with each other?**
- 2. At what time between 2 O' Clock and 3 O' Clock both Hour and Minute hand coincide with each other?**
- 3. At what time between 2 O' Clock and 3 O' Clock both Hour and Minute hand are perpendicular to each other?**
- 4. At what time between 2 O' Clock and 3 O' Clock both Hour and Minute hand are opposite to each other?**
- 5. A person left home between 2 PM and 3 PM and returned home between 4 PM and 5 PM and found that his Hour and Minute hand are changed their places. When did he left the home?**

IDENTITIES**Formulae:**

1. $(a+b)^2 = a^2 + 2ab + b^2 = (a-b)^2 + 4ab$
2. $(a-b)^2 = a^2 - 2ab + b^2 = (a+b)^2 - 4ab$
3. $(a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$
4. $(a+b)^2 - (a-b)^2 = 4ab$
5. $ab = \left(\frac{a+b}{2}\right)^2 - \left(\frac{a-b}{2}\right)^2$
6. $a^2 + b^2 = (a+b)^2 - 2ab = (a-b)^2 + 2ab$
7. $a^2 - b^2 = (a+b)(a-b)$
8. $(x+a)(x+b) = x^2 + x(a+b) + ab$
9. $(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$
10. $a^2 + b^2 + c^2 = (a+b+c)^2 - 2(ab+bc+ca)$
11. $2(ab+bc+ca) = (a+b+c)^2 - (a^2 + b^2 + c^2)$
12. $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3 = a^3 + b^3 + 3ab(a+b)$
13. $(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3 = a^3 - b^3 - 3ab(a-b)$
14. $a^3 + b^3 = (a+b)^3 - 3ab(a+b)$
15. $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$
16. $a^3 - b^3 = (a-b)^3 + 3ab(a-b)$
17. $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$
18. $(a+b)^3 + (a-b)^3 = 2a^3 + 6ab^2 = 2a(a^2 + 3b^2)$
19. $(a+b)^3 - (a-b)^3 = 6a^2b + 2b^3 = 2b(3a^2 + b^2)$
20. $a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$
21. $a^3 + b^3 + c^3 - 3abc = \frac{1}{2}(a+b+c)\{(a-b)^2 + (b-c)^2 + (c-a)^2\}$
22. If $a+b+c=0$, then $a^3 + b^3 + c^3 = 3abc$.
23. $a^4 + a^2b^2 + b^4 = (a^2 + ab + b^2)(a^2 - ab + b^2)$
24. $a^4 - b^4 = (a^2 + b^2)(a+b)(a-b)$
25. $a^n - 1 = (a-1)(1 + a + a^2 + \dots + a^{n-1})$
26. $a^n - b^n = (a-b)(a^{n-1} + a^{n-2}b + a^{n-3}b^2 + \dots + b^{n-1})$

Q. If $a + b + c = 0$, then prove that $a^2 + b^2 + c^2 = -2(ab + bc + ca)$ or $\sum a^2 = -2\sum ab$.

$$\begin{aligned}\text{Ans: } a + b + c = 0 &\Rightarrow (a + b + c)^2 = 0 \Rightarrow a^2 + b^2 + c^2 + 2(ab + bc + ca) = 0 \\ &\Rightarrow a^2 + b^2 + c^2 = -2(ab + bc + ca) \Rightarrow \sum a^2 = -2\sum ab \quad (\text{Proved})\end{aligned}$$

Q. If $a + b + c = 0$, then prove that $a^3 + b^3 + c^3 = 3abc$ or $\sum a^3 = 3abc$.

$$\begin{aligned}\text{Ans: } a + b + c = 0 &\Rightarrow a + b = -c \Rightarrow (a + b)^3 = (-c)^3 \\ &\Rightarrow a^3 + b^3 + 3ab(a + b) = -c^3 \Rightarrow a^3 + b^3 + 3ab(-c) = -c^3 \\ &\Rightarrow a^3 + b^3 + c^3 = 3abc \Rightarrow \sum a^3 = 3abc \quad (\text{Proved})\end{aligned}$$

Q. If $a + b + c = 0$, then prove that $a^4 + b^4 + c^4 = 2(b^2c^2 + c^2a^2 + a^2b^2) = \frac{1}{2}(a^2 + b^2 + c^2)^2$ or.

$$\sum a^4 = 2\sum a^2b^2 = \frac{1}{2}(\sum a^2)^2$$

$$\begin{aligned}\text{Ans: } a + b + c = 0 &\Rightarrow a^2 + b^2 + c^2 = -2(ab + bc + ca) \\ &\Rightarrow (a^2 + b^2 + c^2)^2 = \{-2(ab + bc + ca)\}^2 \\ &\Rightarrow (a^2 + b^2 + c^2)^2 = 4(a^2b^2 + b^2c^2 + c^2a^2 + 2ab^2c + 2abc^2 + 2a^2bc) \\ &\Rightarrow (a^2 + b^2 + c^2)^2 = 4(a^2b^2 + b^2c^2 + c^2a^2 + 2abc(a + b + c)) \\ &\Rightarrow (a^2 + b^2 + c^2)^2 = 4(a^2b^2 + b^2c^2 + c^2a^2 + 2abc(0)) \\ &\Rightarrow (a^2 + b^2 + c^2)^2 = 4(a^2b^2 + b^2c^2 + c^2a^2) \\ &\Rightarrow \frac{1}{2}(a^2 + b^2 + c^2)^2 = 2(a^2b^2 + b^2c^2 + c^2a^2) \text{-----(1)}\end{aligned}$$

$$\begin{aligned}\text{Again form } &\Rightarrow (a^2 + b^2 + c^2)^2 = 4(a^2b^2 + b^2c^2 + c^2a^2) \\ &\Rightarrow a^4 + b^4 + c^4 + 2(a^2b^2 + b^2c^2 + c^2a^2) = 4(a^2b^2 + b^2c^2 + c^2a^2) \\ &\Rightarrow a^4 + b^4 + c^4 = 2(a^2b^2 + b^2c^2 + c^2a^2) \text{-----(2)}\end{aligned}$$

From (1) and (2) $a^4 + b^4 + c^4 = 2(b^2c^2 + c^2a^2 + a^2b^2) = \frac{1}{2}(a^2 + b^2 + c^2)^2$ (proved).

Q. If $a + b + c = 0$, then prove that

$$a^5 + b^5 + c^5 = -5abc(ab + bc + ca) = \frac{5}{2}abc(a^2 + b^2 + c^2) = \frac{5}{6}(a^3 + b^3 + c^3)(a^2 + b^2 + c^2)$$

$$\text{or } \sum a^5 = -5abc \sum ab = \frac{5}{2}abc \sum a^2 = \frac{5}{6} \sum a^3 \sum a^2.$$

$$\begin{aligned}\text{Ans: } a + b + c = 0 &\Rightarrow a + b = -c \Rightarrow (a + b)^5 = (-c)^5 \\ &\Rightarrow a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + b^5 = -c^5 \\ &\Rightarrow a^5 + b^5 + c^5 = -5ab(a^3 + 2a^2b + 2ab^2 + b^3) \\ &\Rightarrow a^5 + b^5 + c^5 = -5ab((a + b)^3 - a^2b - ab^2) \\ &\Rightarrow a^5 + b^5 + c^5 = -5ab((a + b)^3 - ab(a + b))\end{aligned}$$

$$\begin{aligned}
&\Rightarrow a^5 + b^5 + c^5 = -5ab(a+b)((a+b)^2 - ab) \\
&\Rightarrow a^5 + b^5 + c^5 = -5ab(-c)((a+b)(-c) - ab) \\
&\Rightarrow a^5 + b^5 + c^5 = 5abc(-ac - bc - ab) \\
&\Rightarrow a^5 + b^5 + c^5 = -5abc(ab + bc + ca) \text{ (Proved)-----(1)}
\end{aligned}$$

As we know if $a + b + c = 0$, then

$$a^2 + b^2 + c^2 = -2(ab + bc + ca) \Rightarrow (ab + bc + ca) = -\frac{1}{2}(a^2 + b^2 + c^2)$$

By putting this in (1) we get

$$\Rightarrow a^5 + b^5 + c^5 = \frac{5}{2}abc(a^2 + b^2 + c^2) \text{ (Proved)-----(2)}$$

Again we know if $a + b + c = 0$, then $a^3 + b^3 + c^3 = 3abc \Rightarrow abc = \frac{1}{3}(a^3 + b^3 + c^3)$

So equation (2) becomes

$$\Rightarrow a^5 + b^5 + c^5 = \frac{5}{6}(a^3 + b^3 + c^3)(a^2 + b^2 + c^2) \text{ (proved)-----(3)}$$

Exercise:

1. If $a + b + c = 0$ prove that $a^7 + b^7 + c^7 = 7abc(ab + bc + ca)^2$
2. If $a + b + c = 0$ prove that $a^7 + b^7 + c^7 = \frac{7}{4}abc(a^2 + b^2 + c^2)^2$
3. If $a + b + c = 0$ prove that $a^7 + b^7 + c^7 = \frac{7}{3}(a^3 + b^3 + c^3)(ab + bc + ca)^2$
4. If $a + b + c = 0$ prove that $a^7 + b^7 + c^7 = \frac{7}{12}(a^3 + b^3 + c^3)(a^2 + b^2 + c^2)^2$
5. If $a + b + c = 0$ prove that $a^7 + b^7 + c^7 = \frac{7}{6}(a^3 + b^3 + c^3)(a^4 + b^4 + c^4)$
6. If $a + b + c = 0$ prove that $a^7 + b^7 + c^7 = \frac{7}{2}abc(a^4 + b^4 + c^4)$
7. Prove that $(a - b)^3 + (b - c)^3 + (c - a)^3 = 3(a - b)(b - c)(c - a)$.
8. Prove that $(a - b)^5 + (b - c)^5 + (c - a)^5 = \frac{5}{6}((a - b)^3 + (b - c)^3 + (c - a)^3)((a - b)^2 + (b - c)^2 + (c - a)^2)$.
9. If $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{1}{a + b + c}$, then Prove that $\frac{1}{a^{2n+1}} + \frac{1}{b^{2n+1}} + \frac{1}{c^{2n+1}} = \frac{1}{(a + b + c)^{2n+1}}$.
10. If $\frac{a}{b - c} + \frac{b}{c - a} + \frac{c}{a - b} = 0$, then prove that $\frac{a}{(b - c)^2} + \frac{b}{(c - a)^2} + \frac{c}{(a - b)^2} = 0$.

Polynomial

Polynomial: Mathematical expression of the form $a_0 + a_1x + a_2x^2 + \dots + a_nx^n$ is called as a polynomial where $a_0, a_1, a_2, \dots, a_n$ are real numbers, n is a whole number and if $a_n \neq 0$, the polynomial is called as a polynomial of degree n . It is generally denoted as $P(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$

Constant Polynomial: If degree of the polynomial is zero it is called as constant polynomial. It is generally denoted as $P(x) = a_0$, where a_0 are is number.

Linear Polynomial: If degree of the polynomial is one it is called as linear polynomial. It is generally denoted as $P(x) = a_0 + a_1x$, where a_0, a_1 are real numbers.

Quadratic Polynomial: If degree of the polynomial is two it is called as linear polynomial. It is generally denoted as $P(x) = a_0 + a_1x + a_2x^2$, where a_0, a_1, a_2 are real numbers.

Cubic Polynomial: : If degree of the polynomial is three it is called as linear polynomial. It is generally denoted as $P(x) = a_0 + a_1x + a_2x^2 + a_3x^3$, where a_0, a_1, a_2, a_3 are real numbers.

Zeros / Roots of Polynomial: For any value of x , if $P(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n = 0$, then that value is called as zeros or roots of the polynomial.

Remainder Theorem: If any polynomial $P(x)$ is divided by $(x - a)$, then remainder will be $P(a)$.

Factor Theorem: If for any value of $x = a$, $P(a) = 0$, then $(x - a)$ is a factor of $P(x)$.

Q. Find remainder if $3x^2 - 2x + 7$ is divided by $(x - 2)$.

Ans: Here $P(x) = 3x^2 - 2x + 7$, when divided by $(x - 2)$

$$\text{Remainder} = P(2) = 3 \times 2^2 - 2 \times 2 + 7 = 12 - 4 + 7 = 15.$$

Q. Find whether $5x^3 + 3x^2 - x - 7$ has a factor $(x - 1)$

Ans: Here $P(x) = 5x^3 + 3x^2 - x - 7$

$$P(1) = 5 \times 1^3 + 3 \times 1^2 - 1 - 7 = 5 + 3 - 1 - 7 = 0$$

So, by factor theorem $(x - 1)$ is a factor of $5x^3 + 3x^2 - x - 7$.

Q. Find the remainder when $P(x) = x^4 - 5x^3 + 3x^2 - x - 7$ is divided by $(x - 2)(x - 1)$.

Ans: When $P(x) = x^4 - 5x^3 + 3x^2 - x - 7$ is divided by $(x - 2)(x - 1)$ remainder will be $ax + b$.

When $P(x) = x^4 - 5x^3 + 3x^2 - x - 7$ is divided by $(x - 2)$ remainder will be $P(2) = 2a + b$.

When $P(x) = x^4 - 5x^3 + 3x^2 - x - 7$ is divided by $(x - 1)$ remainder will be $P(1) = a + b$.

$$\text{Hence } 2^4 - 5 \times 2^3 + 3 \times 2^2 - 2 - 7 = 2a + b \Rightarrow 2a + b = -20 \text{ -----(1)}$$

$$\text{And } 1^4 - 5 \times 1^3 + 3 \times 1^2 - 1 - 7 = a + b \Rightarrow a + b = -10 \text{ -----(2)}$$

By solving (1) and (2) $a = -10$, $b = 0$

Hence Remainder = $ax + b = -10x - 0 = -10x$ (Ans).

Equation:- An expression of equality is known as equation or an expression equals to another expression is known as equation.

Example: -

- a. $x^4 - 2x^3 + 4x^2 + 6x - 21 = 0$
- b. $2x^4 - 15x^3 + 35x^2 - 30x + 8 = 0$

Algebraic Equation: - If any equation involves polynomial in one variable, then it is called as algebraic equation.

Example: -

- a. $a_0 + a_1x + a_2x^2 + \dots + a_nx^n = 0$
- b. $x^3 - 7x^2 + 36 = 0$

Transcendental Equation: - If any equation involves trigonometric, logarithmic, exponential expression, then it is called as transcendental equation.

Example: -

- a. $\sin x + 3 = 0$
- b. $e^{2x+3} - 36 = 0$
- c. $\log x - 7x^2 + 36 = 0$

Different Types of Algebraic Equation: -

1. **Linear Equation**: - An equation which contains linear polynomial is called as linear equation.
Example: - $ax + b = 0$
2. **Quadratic Equation**: - An equation which contains quadratic polynomial is called as quadratic equation.
Example: - $ax^2 + bx + c = 0$
3. **Cubic Equation**: - An equation which contains cubic polynomial is called as cubic equation.
Example: - $ax^3 + bx^2 + cx + d = 0$
4. **Biquadratic Equation**: - An equation which contains biquadratic polynomial is called as biquadratic equation.
Example: - $ax^4 + bx^3 + cx^2 + dx + e = 0$

Solution of Algebraic Equation: - For any value of x which satisfies $f(x) = 0$ are called as solution of $f(x) = 0$ or root of $f(x) = 0$.

Example: - For $2x + 6 = 0$

- a. If $x = 2$, it is not satisfies $2x + 6 = 0$. So, $x = 2$ is not a solution of $2x + 6 = 0$.
- b. If $x = -3$, it is not satisfies $2x + 6 = 0$. So, $x = -3$ is a solution of $2x + 6 = 0$.

General Properties of roots of Equation: -

1. If α is a root of the equation $f(x) = 0$, then $f(x)$ is exactly divisible by $x - \alpha$ or $x - \alpha$ is a factor of $f(x)$.
2. If $f(x)$ is exactly divisible by $x - \alpha$, then α is a root of $f(x) = 0$.
3. Every equation of n th degree has exactly n roots, may be real, and may be imaginary.
4. Let $f(x) = 0$ such that, $f(a)$ and $f(b)$ have different sign then equation $f(x) = 0$ has at least one root lies between a and b .

5. In an equation with real coefficients, if $\alpha + i\beta$ is a root then $\alpha - i\beta$ is also one of the root.
Or imaginary roots always occur in conjugate pairs.
6. If $f(x) = 0$ has $a + \sqrt{b}$ is one of the root, then $a - \sqrt{b}$ is also another root.
7. Every equation of odd degree has atleast one real root.

Descarte's Rule of Sign:-

- a. Maximum number of positive real root of $f(x) = 0$ is same as the number of sign changes in $f(x)$.
- b. Maximum number of negative real root of $f(x) = 0$ is same as the number of sign changes in $f(-x)$.
- c. If an equation having degree n such that at most p number of positive roots, at most q number of negative roots, then equation has atleast $n - (p + q)$ number of imaginary roots.

Example: -

1. Find the number of positive, negative and imaginary roots of the equation $x^6 - x^5 - 10x + 7 = 0$.

Ans: - Number of Positive roots = 2
 Number of Negative roots = 0
 Number of Imaginary roots = 4.

Relation between Roots and Coefficients of an equation: -

- a. For the quadratic equation $a_0x^2 + a_1x + a_2 = 0$. If α_1, α_2 are two roots then

$$\sum \alpha_i = \alpha_1 + \alpha_2 = -\frac{a_1}{a_0}$$

$$\alpha_1 \cdot \alpha_2 = \frac{a_2}{a_0}$$

- b. For the cubic equation $a_0x^3 + a_1x^2 + a_2x + a_3 = 0$. If $\alpha_1, \alpha_2, \alpha_3$ are three roots then

$$\sum \alpha_i = \alpha_1 + \alpha_2 + \alpha_3 = -\frac{a_1}{a_0}$$

$$\sum \alpha_i \alpha_j = \alpha_1 \cdot \alpha_2 + \alpha_2 \cdot \alpha_3 + \alpha_3 \cdot \alpha_1 = \frac{a_2}{a_0}$$

$$\alpha_1 \cdot \alpha_2 \cdot \alpha_3 = -\frac{a_3}{a_0}$$

- c. Similarly, Let us consider n th degree equation $a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_n = 0$. Let $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n$ are n numbers of roots then

$$\sum \alpha_i = -\frac{a_1}{a_0}$$

$$\sum \alpha_i \alpha_j = \frac{a_2}{a_0}$$

$$\sum \alpha_i \alpha_j \alpha_k = -\frac{a_3}{a_0} \dots \dots \dots$$

$$\alpha_1 \alpha_2 \alpha_3 \dots \alpha_n = (-1)^n \frac{a_n}{a_0}$$

Example: -

1. If α, β, γ are the roots of the equation $x^3 + px + q = 0$. Find the value of:

- i.** $\sum \alpha^2 \beta$.
- ii.** $\sum \alpha^4$.
- iii.** $\sum \alpha^3 \beta$.

Ans: - α, β, γ are the roots of the equation $x^3 + px + q = 0$

$$\text{Here } \sum \alpha = 0, \sum \alpha\beta = p, \alpha\beta\gamma = -q$$

$$(i) \sum \alpha^2 \beta = \alpha^2 \beta + \alpha^2 \gamma + \beta^2 \alpha + \beta^2 \gamma + \gamma^2 \alpha + \gamma^2 \beta$$

$$\text{Now } (\alpha + \beta + \gamma)(\alpha\beta + \beta\gamma + \gamma\alpha) = 0 \times p = 0$$

$$\Rightarrow \alpha^2 \beta + \alpha^2 \gamma + \beta^2 \alpha + \beta^2 \gamma + \gamma^2 \alpha + \gamma^2 \beta + 3\alpha\beta\gamma = 0$$

$$\Rightarrow \sum \alpha^2 \beta + 3\alpha\beta\gamma = 0$$

$$\Rightarrow \sum \alpha^2 \beta + 3(-q) = 0$$

$$\Rightarrow \sum \alpha^2 \beta = 3q$$

$$(ii) \sum \alpha^4 = \alpha^4 + \beta^4 + \gamma^4$$

$$x^3 + px + q = 0$$

$$\Rightarrow x^4 + px^2 + qx = 0$$

$$\Rightarrow \sum \alpha^4 + p \sum \alpha^2 + q \sum \alpha = 0$$

$$\Rightarrow \sum \alpha^4 + p \left\{ \left(\sum \alpha \right)^2 - 2 \sum \alpha\beta \right\} + q \sum \alpha = 0$$

$$\Rightarrow \sum \alpha^4 + p \{0 - 2p\} + q \cdot 0 = 0$$

$$\Rightarrow \sum \alpha^4 = 2pq. \text{ (Ans)}$$

$$(iii) \sum \alpha^3 \beta = \alpha^3 \beta + \alpha^3 \gamma + \beta^3 \alpha + \beta^3 \gamma + \gamma^3 \alpha + \gamma^3 \beta$$

$$\text{Now } (\alpha^2 + \beta^2 + \gamma^2)(\alpha\beta + \beta\gamma + \gamma\alpha) = \{(\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)\} \times p$$

$$\Rightarrow \alpha^3 \beta + \alpha^3 \gamma + \beta^3 \alpha + \beta^3 \gamma + \gamma^3 \alpha + \gamma^3 \beta + \alpha^2 \beta\gamma + \beta^2 \gamma\alpha + \gamma^2 \alpha\beta = \{0 - 2p\} \times p$$

$$\Rightarrow \sum \alpha^3 \beta + \alpha\beta\gamma(\alpha + \beta + \gamma) = -2p^2$$

$$\Rightarrow \sum \alpha^3 \beta + 0 = -2p^2$$

$$\Rightarrow \sum \alpha^3 \beta = -2p^2. \text{ (Ans)}$$

2. If α, β, γ are the roots of the equation $x^3 + px + q = 0$. Show that

- i.** $\alpha^5 + \beta^5 + \gamma^5 = -5\alpha\beta\gamma \sum \alpha\beta$.
- ii.** $3 \sum \alpha^2 \sum \alpha^5 = 5 \sum \alpha^3 \sum \alpha^4$.

Ans: As α, β, γ are the roots of the cubic $x^3 + px + q = 0$.-----(1)

$$\text{Here } \sum \alpha = 0, \sum \alpha\beta = p, \alpha\beta\gamma = -q$$

$$\text{Now, } \sum \alpha^2 = \left(\sum \alpha \right)^2 - 2 \sum \alpha\beta = 0 - 2p = -2p.$$

$$\Rightarrow \sum \alpha^2 = -2p \text{ -----(2)}$$

$$\text{As } \sum \alpha = 0 \Rightarrow \sum \alpha^3 - 3\alpha\beta\gamma = 0 \Rightarrow \sum \alpha^3 - 3(-q) = 0$$

$$\Rightarrow \sum \alpha^3 = -3q \text{ -----(3)}$$

Let us multiply equation (1) by x we get

$$\Rightarrow x^4 + px^2 + qx = 0$$

$$\Rightarrow \sum \alpha^4 + p \sum \alpha^2 + q \sum \alpha = 0$$

$$\Rightarrow \sum \alpha^4 = -p \sum \alpha^2 - q \sum \alpha = -p(-2p) - q(0) = 2p^2$$

$$\Rightarrow \sum \alpha^4 = 2p^2 \text{ -----(3)}$$

i. $\alpha^5 + \beta^5 + \gamma^5 = -5\alpha\beta\gamma \sum \alpha\beta$

Ans: Now by multiplying (1) by x^2 we get $\Rightarrow x^5 + px^3 + qx^2 = 0$

$$\Rightarrow \sum \alpha^5 + p \sum \alpha^3 + q \sum \alpha^2 = 0$$

$$\Rightarrow \sum \alpha^5 = -p \sum \alpha^3 - q \sum \alpha^2 = -p(-3q) - q(-2p) = 5pq$$

$$\Rightarrow \sum \alpha^5 = 5pq \text{ -----(4)}$$

$$\Rightarrow \sum \alpha^5 = -5(-q)p$$

$$\Rightarrow \sum \alpha^5 = -5\alpha\beta\gamma \sum \alpha\beta$$

$$\Rightarrow \alpha^5 + \beta^5 + \gamma^5 = -5\alpha\beta\gamma \sum \alpha\beta \text{ (Proved)}$$

ii. $3 \sum \alpha^2 \sum \alpha^5 = 5 \sum \alpha^3 \sum \alpha^4$

Ans: $3 \sum \alpha^2 \sum \alpha^5 = 3(-2p)(5pq) = -30p^2q \text{ -----(5)}$

$$5 \sum \alpha^3 \sum \alpha^4 = 5(-3q)(2p^2) = -30p^2q \text{ -----(6)}$$

From (5) and (6) $\Rightarrow 3 \sum \alpha^2 \sum \alpha^5 = 5 \sum \alpha^3 \sum \alpha^4 \text{ (Proved)}$

How to find common roots of the pair of equations:

Q. Find the common roots of the pair of equations

$$6x^3 - 17x^2 + 11x - 2 = 0 \quad \text{and} \quad 12x^3 - 4x^2 - 3x + 1 = 0$$

$$\begin{array}{r}
 2 \\
 6x^3 - 17x^2 + 11x - 2 \quad \left| \begin{array}{l} 12x^3 - 4x^2 - 3x + 1 \\ 12x^3 - 34x^2 + 22x - 4 \end{array} \right. \\
 \hline
 30x^2 - 25x + 5 \quad x \\
 6x^2 - 5x + 1 \quad \left| \begin{array}{l} 6x^3 - 17x^2 + 11x - 2 \\ 6x^3 - 5x^2 + x \end{array} \right. \\
 \hline
 -12x^2 + 10x - 2 \quad 1 \\
 6x^2 - 5x + 1 \quad \left| \begin{array}{l} 6x^2 - 5x + 1 \\ 6x^2 - 5x + 1 \end{array} \right. \\
 \hline
 0
 \end{array}$$

By division method HCF = $6x^2 - 5x + 1$

If $6x^2 - 5x + 1 = 0$, then $(2x-1)(3x-1) = 0 \Rightarrow x = \frac{1}{2}, \frac{1}{3}$

So common roots are $\frac{1}{2}, \frac{1}{3}$.

Exercise:

1. Solve $x^3 - 7x^2 + 36 = 0$, where one root is double of other.
2. Find the condition that cubic $x^3 - lx^2 + mx - n = 0$ should have its roots in A.P.
3. Find the condition that cubic $x^3 - lx^2 + mx - n = 0$ should have its roots in G.P.
4. Solve $x^4 - 2x^3 - 21x^2 + 22x + 40 = 0$, whose roots are in A.P.
5. Solve $2x^4 - 15x^3 + 35x^2 - 30x + 8 = 0$, whose roots are in G.P.
6. Solve $x^4 - 2x^3 - 22x^2 + 62x - 15 = 0$, whose one of the root is $2 + \sqrt{3}$.
7. The equation $x^4 - 4x^3 + ax^2 + 4x + b = 0$ has two pairs of equal roots. Find the value of a & b.
8. Solve $x^3 - 9x^2 + 14x + 24 = 0$, given that two of its roots are in ratio 3:2.
9. Solve $x^3 - 4x^2 - 20x + 48 = 0$, given that roots α, β are roots connected by $\alpha + 2\beta = 0$.
10. Solve $x^3 - 12x^2 + 39x - 28 = 0$, whose roots are in A.P.
11. Solve $8x^3 - 14x^2 + 7x - 1 = 0$, whose roots are in G.P.
12. O, A, B, C are four points on a straight line such that the distance of A, B, C from O are roots of equation $ax^3 + 3bx^2 + 3cx + d = 0$,. If B is the mid point of AC show that $a^2d - 3abc + 2b^3 = 0$.
13. If α, β, γ are the roots of cubic equation $x^3 - lx^2 + mx - n = 0$, find $\sum \alpha^2 \beta^2$.
14. If α, β, γ are the roots of cubic equation $x^3 - lx^2 + mx - n = 0$, find $(\beta + \gamma)(\gamma + \alpha)(\alpha + \beta)$.
15. Find sum of cubes of roots of the equation $x^3 - 6x^2 + 11x - 6 = 0$.
16. If α, β, γ are the roots of cubic equation $x^3 + 4x - 3 = 0$. Find the value of $\alpha^{-1} + \beta^{-1} + \gamma^{-1}$.
17. If α, β, γ are the roots of cubic equation $x^3 + 4x - 3 = 0$. Find the value of $\alpha^{-2} + \beta^{-2} + \gamma^{-2}$.
18. If α, β, γ are the roots of cubic equation $x^3 + px + q = 0$. Show that $\alpha^5 + \beta^5 + \gamma^5 = 5\alpha\beta\gamma(\alpha\beta + \beta\gamma + \alpha\gamma)$.
19. Find common roots between $x^4 + 5x^3 - 22x^2 - 50x + 132 = 0$ and $x^4 + x^3 - 20x^2 + 16x + 24 = 0$
20. Find common roots between $3x^3 + 17x^2 - 62x + 14 = 0$ and $7x^3 + 52x^2 - 46x + 8 = 0$

Inequality

Absolute Value Function: If x be any real number, then the absolute value of x , written as $|x|$ is defined as follows :

$$|x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

Properties : For any real numbers x, y

1. $|x| = |-x|$
2. $|x|^2 = x^2$
3. $|x| = \max\{x, -x\}$
4. $|x + y| \leq |x| + |y|$
5. $|xy| = |x||y|$
6. $|x - y| \geq ||x| - |y||$

Integral Part of a number (greatest Integer): The integral part of a number x is denoted by $[x]$ and is also called as greatest integer of x .

$[x] = x$ if x is an integer.

$[x]$ = nearest left side integer of x if x is not an integer.

In other word $[x] \leq x$.

Example: $[5] = 5$, $[-23] = -23$, $[100] = 100$ etc as in each case x is an integer.

$[5.9] = 5$, $[5.001] = 5$, $[-15.1] = -16$, $[103.33] = 103$ etc.

Properties: 1. $[x] \leq x < [x] + 1$, Where $[x]$ is the greatest integer of x .

2. $[Nx] \leq Nx \leq [Nx] + 1$

Q. Find the integral part of $1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \frac{1}{\sqrt{4}} + \frac{1}{\sqrt{5}}$.

Ans: Here $1 \leq 1 \leq 1$, $0.7 < \frac{1}{\sqrt{2}} < 0.8$, $0.5 < \frac{1}{\sqrt{3}} < 0.6$, $0.5 \leq \frac{1}{\sqrt{4}} \leq 0.5$, $0.4 < \frac{1}{\sqrt{5}} < 0.5$.

Hence $1 + 0.7 + 0.5 + 0.5 + 0.4 < 1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \frac{1}{\sqrt{4}} + \frac{1}{\sqrt{5}} < 1 + 0.8 + 0.6 + 0.5 + 0.5$

$\Rightarrow 3.1 < 1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \frac{1}{\sqrt{4}} + \frac{1}{\sqrt{5}} < 3.4$

Integral part of $1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \frac{1}{\sqrt{4}} + \frac{1}{\sqrt{5}}$ is 3.

Q. Prove that $2\sqrt{n+1} - 2\sqrt{n} < \frac{1}{\sqrt{n}} < 2\sqrt{n} - 2\sqrt{n-1}$

$$\text{Ans: } 2\sqrt{n+1} - 2\sqrt{n} = (2\sqrt{n+1} - 2\sqrt{n}) \times \frac{(2\sqrt{n+1} + 2\sqrt{n})}{(2\sqrt{n+1} + 2\sqrt{n})} = \frac{4(n+1) - 4n}{2(\sqrt{n+1} + \sqrt{n})} = \frac{4}{2(\sqrt{n+1} + \sqrt{n})}$$

$$\Rightarrow 2\sqrt{n+1} - 2\sqrt{n} = \frac{2}{\sqrt{n+1} + \sqrt{n}} < \frac{2}{\sqrt{n} + \sqrt{n}} = \frac{2}{2\sqrt{n}} = \frac{1}{\sqrt{n}} \text{-----(1)}$$

$$2\sqrt{n} - 2\sqrt{n-1} = (2\sqrt{n} - 2\sqrt{n-1}) \times \frac{(2\sqrt{n} + 2\sqrt{n-1})}{(2\sqrt{n} + 2\sqrt{n-1})} = \frac{4n - 4(n-1)}{2(\sqrt{n} + \sqrt{n-1})} = \frac{4}{2(\sqrt{n} + \sqrt{n-1})}$$

$$\Rightarrow 2\sqrt{n} - 2\sqrt{n-1} = \frac{2}{\sqrt{n} + \sqrt{n-1}} > \frac{2}{\sqrt{n} + \sqrt{n}} = \frac{2}{2\sqrt{n}} = \frac{1}{\sqrt{n}} \text{-----(2)}$$

From (1) and (2) $2\sqrt{n+1} - 2\sqrt{n} < \frac{1}{\sqrt{n}} < 2\sqrt{n} - 2\sqrt{n-1}$ (Proved)

Q. Prove that $2\sqrt{k} - 2 < 1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{k}} < 2\sqrt{k} - 1$.

Ans: As per the above proof $2\sqrt{n+1} - 2\sqrt{n} < \frac{1}{\sqrt{n}} < 2\sqrt{n} - 2\sqrt{n-1}$

By putting $n = 2, 3, 4, \dots, k$ we get

$$2\sqrt{3} - 2\sqrt{2} < \frac{1}{\sqrt{2}} < 2\sqrt{2} - 2$$

$$2\sqrt{4} - 2\sqrt{3} < \frac{1}{\sqrt{3}} < 2\sqrt{3} - 2\sqrt{2}$$

$$2\sqrt{5} - 2\sqrt{4} < \frac{1}{\sqrt{4}} < 2\sqrt{4} - 2\sqrt{3}$$

.....

$$2\sqrt{k+1} - 2\sqrt{k} < \frac{1}{\sqrt{k}} < 2\sqrt{k} - 2\sqrt{k-1}$$

By adding all these we get

$$2\sqrt{k+1} - 2\sqrt{2} < \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{k}} < 2\sqrt{k} - 2$$

$$\Rightarrow 2\sqrt{k+1} - 2\sqrt{2} + 1 < 1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{k}} < 2\sqrt{k} - 2 + 1 \quad (\text{as } 2\sqrt{2} < 3 \text{ and } \sqrt{k+1} < \sqrt{k})$$

$$\Rightarrow 2\sqrt{k} - 3 + 1 < 1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{k}} < 2\sqrt{k} - 2 + 1$$

$$\Rightarrow 2\sqrt{k} - 2 < 1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{k}} < 2\sqrt{k} - 1$$

Q. Find integral part of $1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{1000000}}$

Ans: As per the above proof $2\sqrt{k} - 2 < 1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{k}} < 2\sqrt{k} - 1$.

Now putting $k = 1000000$

$$\Rightarrow 2\sqrt{1000000} - 2 < 1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{1000000}} < 2\sqrt{1000000} - 1$$

$$\Rightarrow 1998 < 1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{1000000}} < 1999$$

Hence integral part of $1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{1000000}}$ is 1998.

Q. Prove that $\frac{1}{2} \times \frac{3}{4} \times \frac{5}{6} \times \dots \times \frac{99}{100} < \frac{1}{10}$

Ans: Let $x = \frac{1}{2} \times \frac{3}{4} \times \frac{5}{6} \times \dots \times \frac{99}{100}$ and $y = \frac{2}{3} \times \frac{4}{5} \times \frac{6}{7} \times \dots \times \frac{100}{101}$

As $\frac{1}{2} < \frac{2}{3}, \frac{3}{4} < \frac{4}{5}, \frac{5}{6} < \frac{6}{7}, \dots, \frac{99}{100} < \frac{100}{101}$, so $x < y$

$$\Rightarrow x^2 < xy = \frac{1}{2} \times \frac{3}{4} \times \frac{5}{6} \times \dots \times \frac{99}{100} \times \frac{2}{3} \times \frac{4}{5} \times \frac{6}{7} \times \dots \times \frac{100}{101}$$

$$\Rightarrow x^2 < \frac{1}{101} \Rightarrow x < \frac{1}{\sqrt{101}} < \frac{1}{\sqrt{100}} = \frac{1}{10}. \text{ So, } \frac{1}{2} \times \frac{3}{4} \times \frac{5}{6} \times \dots \times \frac{99}{100} < \frac{1}{10} \text{ (Proved).}$$

Exercise:

1. Prove that $\frac{1}{2} \times \frac{3}{4} \times \frac{5}{6} \times \dots \times \frac{99}{100} < \frac{1}{12}$.

2. Prove that $\frac{1}{2} \times \frac{3}{4} \times \frac{5}{6} \times \dots \times \frac{2n-1}{2n} < \frac{1}{\sqrt{3n+1}}$.

3. Prove that $2\sqrt{n+1} - 2\sqrt{m} < \frac{1}{\sqrt{m}} + \frac{1}{\sqrt{m+1}} + \dots + \frac{1}{\sqrt{n}} < 2\sqrt{n} - 2\sqrt{m-1}$.

4. Find integral part of $\frac{1}{\sqrt{10000}} + \frac{1}{\sqrt{10001}} + \dots + \frac{1}{\sqrt{1000000}}$.

5. Find integral part of $50x$, where $x = \frac{1}{\sqrt{10000}} + \frac{1}{\sqrt{10001}} + \dots + \frac{1}{\sqrt{1000000}}$.

Elementary Inequalities

1. If any real number a is given then: either $a < 0$ or $a = 0$ or $a > 0$.
2. If any two real numbers a and b are given then: either $a < b$ or $a = b$ or $a > b$.
3. If $a > 0$ and $b > 0$, then $ab > 0$.
4. If $a > b$, then $a - b > 0$.
5. If $a < b$, then $b > a$.
6. If $a \leq b$, then either $a < b$ or $a = b$.
7. If $a \geq b$, then either $a > b$ or $a = b$.
8. If $a > b$ and $b > c$, then $a > c$.
9. In general if $a_1 \geq a_2, a_2 \geq a_3, a_3 \geq a_4, \dots, a_{n-1} \geq a_n$, then $a_1 \geq a_n$.
10. If $a > b$, then for any real number c , $a + c > b + c$.
11. If $a > b$ and $c > d$, then $a + c > b + d$.
12. In general $a_1 \geq b_1, a_2 \geq b_2, a_3 \geq b_3, \dots, a_n \geq b_n$, then $a_1 + a_2 + \dots + a_n \geq b_1 + b_2 + \dots + b_n$.
13. If $a \geq b$ and $c \geq d$, then $a - d \geq b - c$.
14. If $a > b$, then for any real number $c > 0$, $ac > bc$.
15. If $a > b$, then for any real number $c < 0$, $ac < bc$.
16. If $a > b > 0$ and $c > d > 0$, then $ac > bd > 0$.
17. If $a > b > 0$, then $\frac{1}{a} < \frac{1}{b}$.
18. If $a > b > 0$ and $c > d > 0$, then $\frac{a}{d} > \frac{b}{c}$.
19. $a^2 \geq 0$.

Arithmetic Mean: Arithmetic mean between two numbers a and b is $\frac{a+b}{2}$.

$$\text{Arithmetic mean of } a_1, a_2, \dots, a_n = \frac{a_1 + a_2 + \dots + a_n}{n}.$$

Geometric Mean: Geometric mean between two numbers a and b is $\sqrt{ab}$.

$$\text{Geometric mean of } a_1, a_2, \dots, a_n = (a_1 a_2 \dots a_n)^{\frac{1}{n}}.$$

Harmonic Mean: Harmonic mean between two numbers a and b is $\left(\frac{\frac{1}{a} + \frac{1}{b}}{2} \right)^{-1} = \frac{2ab}{a+b}$.

$$\text{Harmonic mean of } a_1, a_2, \dots, a_n = \left(\frac{\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n}}{n} \right)^{-1}.$$

Properties: 1. $AM \geq GM \geq HM$, Where AM is the arithmetic mean, GM is the geometric mean and HM is the harmonic mean.

$$2. \frac{a+b}{2} \geq \sqrt{ab} \geq \frac{2ab}{a+b}.$$

$$3. \frac{a_1 + a_2 + \dots + a_n}{n} \geq (a_1 a_2 \dots a_n)^{\frac{1}{n}} \geq \left(\frac{\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n}}{n} \right)^{-1}$$

Q. For any positive real number a , prove that $a + \frac{1}{a} \geq 2$.

Proof: Let a is any positive real number, then $\frac{1}{a}$ is also a positive real number.

As $AM \geq GM$

$$\Rightarrow \frac{a + \frac{1}{a}}{2} \geq \sqrt{a \times \frac{1}{a}} \Rightarrow \frac{a + \frac{1}{a}}{2} \geq 1 \Rightarrow a + \frac{1}{a} \geq 2 \text{ (Proved).}$$

$$\text{Or } \left(\sqrt{a} - \frac{1}{\sqrt{a}} \right)^2 \geq 0 \Rightarrow a + \frac{1}{a} - 2 \geq 0 \Rightarrow a + \frac{1}{a} \geq 2 \text{ (Proved).}$$

Q. If a, b, c are three positive real numbers, then prove that $(a + b + c)^3 \geq 27abc$

Ans: As $AM \geq GM$

$$\Rightarrow \frac{a + b + c}{3} \geq (abc)^{\frac{1}{3}} \Rightarrow \frac{(a + b + c)^3}{27} \geq abc \Rightarrow (a + b + c)^3 \geq 27abc \text{ (Proved).}$$

Q. If a, b, c are three positive real numbers, then prove that $(a + b)(b + c)(c + a) \geq 8abc$

Ans: As $AM \geq GM$

$$\frac{a + b}{2} \geq \sqrt{ab} \Rightarrow a + b \geq 2\sqrt{ab}, \text{ similarly } b + c \geq 2\sqrt{bc} \text{ and } c + a \geq 2\sqrt{ca}$$

$$\Rightarrow (a + b)(b + c)(c + a) \geq 2\sqrt{ab} 2\sqrt{bc} 2\sqrt{ca}$$

$$\Rightarrow (a + b)(b + c)(c + a) \geq 8\sqrt{a^2 b^2 c^2}$$

$$\Rightarrow (a + b)(b + c)(c + a) \geq 8abc \text{ (Proved).}$$

Q. Prove that $\left(\frac{n+1}{2} \right)^n \geq n!$, where $n! = 1 \times 2 \times 3 \times 4 \times \dots \times n$

Ans: As $AM \geq GM$

$$\Rightarrow \frac{1 + 2 + 3 + \dots + n}{n} \geq (1 \times 2 \times 3 \times \dots \times n)^{\frac{1}{n}} \Rightarrow \frac{n(n+1)}{2n} \geq (n!)^{\frac{1}{n}} \Rightarrow \frac{(n+1)}{2} \geq (n!)^{\frac{1}{n}} \Rightarrow \left(\frac{n+1}{2} \right)^n \geq n!.$$

Exercise:

1. If $a, b > 0$, then prove that $\frac{a}{b} + \frac{b}{a} \geq 2$.
2. If $a, b > 0$ are positive real numbers, then prove that $(a+b)\left(\frac{1}{a} + \frac{1}{b}\right) \geq 4$.
3. If $a, b, c > 0$, then prove that $\frac{a}{b} + \frac{b}{c} + \frac{c}{a} \geq 3$.
4. If $a, b, c > 0$ and $a+b+c=1$, then prove that $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 9$.
5. If $a, b, c > 0$ and $a+b+c=1$, then prove that $\frac{1}{1+3abc} \geq \frac{9}{10}$.
6. If $a, b, c > 0$ and $a+b+c=1$, then prove that $ab(a+b)^2 + bc(b+c)^2 + ca(c+a)^2 \geq 4abc$.
7. If $a_1, a_2, \dots, a_n > 0$, then prove that $\frac{a_1}{a_2} + \frac{a_2}{a_3} + \frac{a_3}{a_4} + \dots + \frac{a_{n-1}}{a_n} + \frac{a_n}{a_1} \geq n$.
8. For any real number a prove that $\frac{a^2}{a^4+1} \leq \frac{1}{2}$.
9. Prove that $\frac{a^2+2}{\sqrt{a^2+1}} \geq 2$.
10. Prove that $\frac{a+nb}{n+1} \geq (ab^n)^{\frac{1}{n+1}}$.
11. For $a > 1$, prove that $\log a + \log_a 10 \geq 2$.
12. If $a, b, c, d > 0$ then prove that $(ac+bd)(ab+cd) \geq 4abcd$.
13. If $a, b > 0$, then prove that $a^2 + b^2 \geq 2ab$.
14. If $a, b, c > 0$ such that $abc=1$, then prove that $a^3 + b^3 + c^3 \geq 3$.
15. If $a, b, c > 0$, then prove that $a^3 + b^3 + c^3 \geq 3abc$.
16. If $a, b, c, d > 0$, then prove that $a^4 + b^4 + c^4 + d^4 \geq 4abcd$.
17. If $a, b, c, d > 0$, then prove that $\sqrt{(a+c)(b+d)} \geq \sqrt{ab} + \sqrt{cd}$.
18. If $a_1, a_2, \dots, a_n > 0$, then prove that $a_1^n + a_2^n + \dots + a_n^n \geq n.a_1 a_2 \dots a_n$.
19. If $a_1, a_2, \dots, a_n > 0$, then prove that $(a_1 + a_2 + \dots + a_n)\left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n}\right) \geq n^2$.
20. If $a, b, c > 0$, then prove that $(a+b+c)(ab+bc+ca) \geq 9abc$.
21. If $a, b, c > 0$, then prove that $(a^2b + b^2c + c^2a)(ab^2 + bc^2 + ca^2) \geq 9a^2b^2c^2$.
22. Prove that $(n+1)^n \geq 2^n \cdot n!$
23. Prove that $2^n \times n^n \geq \frac{(2n)!}{n!}$.

24. Prove that $\left(\frac{(n+1)(2n+1)}{6}\right)^n \geq (n!)^2$.

25. Prove that $n^n \left(\frac{n+1}{2}\right)^{2n} \geq (n!)^3$.

26. If $a, b, c, d > 0$, then prove that $\frac{1}{a^3} + \frac{1}{b^3} + \frac{1}{c^3} + \frac{1}{d^3} \geq \frac{1}{abc} + \frac{1}{bcd} + \frac{1}{cda} + \frac{1}{dab}$.

Cauchy-Schwarz Inequality:

1. If a, b and x, y are real numbers then $(ax + by)^2 \leq (a^2 + b^2)(x^2 + y^2)$.
2. If a, b, c and x, y, z are real numbers then $(ax + by + cz)^2 \leq (a^2 + b^2 + c^2)(x^2 + y^2 + z^2)$.
3. If a_i, b_i are real numbers for $i = 1, 2, 3, \dots, n$ then

$$(a_1 b_1 + a_2 b_2 + \dots + a_n b_n)^2 \leq (a_1^2 + a_2^2 + \dots + a_n^2)(b_1^2 + b_2^2 + \dots + b_n^2)$$

Q. Prove that If a, b and x, y are real numbers, then $(ax + by)^2 \leq (a^2 + b^2)(x^2 + y^2)$.

Ans: $(a^2 + b^2)(x^2 + y^2) - (ax + by)^2 = a^2 x^2 + a^2 y^2 + b^2 x^2 + b^2 y^2 - a^2 x^2 - b^2 y^2 - 2axby$

$$\Rightarrow (a^2 + b^2)(x^2 + y^2) - (ax + by)^2 = a^2 y^2 + b^2 x^2 - 2axby$$

$$\Rightarrow (a^2 + b^2)(x^2 + y^2) - (ax + by)^2 = a^2 y^2 + b^2 x^2 - 2aybx = (ay + bx)^2 \geq 0$$

$$\Rightarrow (a^2 + b^2)(x^2 + y^2) - (ax + by)^2 \geq 0$$

$$\Rightarrow (a^2 + b^2)(x^2 + y^2) \geq (ax + by)^2 \text{ Proved.}$$

Q. Prove that If a, b, c and x, y, z are real numbers, then $(ax + by + cz)^2 \leq (a^2 + b^2 + c^2)(x^2 + y^2 + z^2)$.

Ans: $(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) - (ax + by + cz)^2 = (ay - bx)^2 + (bz - cy)^2 + (cx - az)^2 \geq 0$

$$\Rightarrow (a^2 + b^2 + c^2)(x^2 + y^2 + z^2) - (ax + by + cz)^2 \geq 0$$

$$\Rightarrow (a^2 + b^2 + c^2)(x^2 + y^2 + z^2) \geq (ax + by + cz)^2 \text{ (Proved)}$$

Q. If x, y, z are real numbers, then prove that $\left(\frac{x}{2} + \frac{y}{3} + \frac{z}{6}\right)^2 \leq \frac{x^2}{2} + \frac{y^2}{3} + \frac{z^2}{6}$.

Ans: By putting Cauchy Schwartz Inequality to $x, x, x, y, y, y, z, z, z$ and $1, 1, 1, 1, 1, 1, 1, 1, 1$ we get

$$\Rightarrow (x^2 + x^2 + x^2 + y^2 + y^2 + y^2 + z^2 + z^2 + z^2)(1^2 + 1^2 + 1^2 + 1^2 + 1^2 + 1^2) \geq (x.1 + x.1 + x.1 + y.1 + y.1 + y.1 + z.1 + z.1 + z.1)^2$$

$$\Rightarrow 6(3x^2 + 2y^2 + z^2) \geq (3x + 2y + z)^2$$

$$\Rightarrow \frac{(3x + 2y + z)^2}{36} \leq \frac{6(3x^2 + 2y^2 + z^2)}{36}$$

$$\Rightarrow \left(\frac{3x + 2y + z}{6}\right)^2 \leq \frac{(3x^2 + 2y^2 + z^2)}{6}$$

$$\Rightarrow \left(\frac{x}{2} + \frac{y}{3} + \frac{z}{6}\right)^2 \leq \frac{x^2}{2} + \frac{y^2}{3} + \frac{z^2}{6} \text{ (Proved).}$$

Exercise:

1. If a, b, x, y are real numbers such that $a^2 + b^2 = 1$ and $x^2 + y^2 = 1$, then prove that $(ax + by) \leq 1$.
2. If a, b, c, x, y, z are real numbers such that $a^2 + b^2 + c^2 = 1$ and $x^2 + y^2 + z^2 = 1$, then prove that $(ax + by + cz) \leq 1$.
3. If $a, b, c > 0$, such that $a^3 + b^3 + c^3 = 81$, then prove that $a + b + c \leq 9$.
4. If $a, b, c > 0$, such that $a^2 + b^2 + c^2 = 27$, then prove that $a^3 + b^3 + c^3 \geq 81$.
5. If $a, b, c > 0$, such that $a^2 + b^2 + c^2 = 16$, then prove that $a^3 + b^3 + c^3 \geq 16\sqrt{\frac{2}{3}}$.

Tchebyschef's Inequality:

1. If $a \leq b$ and $x \leq y$ are real numbers then $2(ax + by) \geq (a + b)(x + y)$.
2. If $a \geq b$ and $x \leq y$ are real numbers then $2(ax + by) \leq (a + b)(x + y)$.
3. If $a \leq b \leq c$ and $x \leq y \leq z$ are real numbers then $3(ax + by + cz) \geq (a + b + c)(x + y + z)$.
4. If $a \geq b \geq c$ and $x \leq y \leq z$ are real numbers then $3(ax + by + cz) \leq (a + b + c)(x + y + z)$.
5. If $a_1 \leq a_2 \leq \dots \leq a_n$ and $b_1 \leq b_2 \leq \dots \leq b_n$ are real numbers then

$$n(a_1 b_1 + a_2 b_2 + \dots + a_n b_n) \geq (a_1 + a_2 + \dots + a_n)(b_1 + b_2 + \dots + b_n)$$
6. If $a_1 \geq a_2 \geq \dots \geq a_n$ and $b_1 \leq b_2 \leq \dots \leq b_n$ are real numbers then

$$n(a_1 b_1 + a_2 b_2 + \dots + a_n b_n) \leq (a_1 + a_2 + \dots + a_n)(b_1 + b_2 + \dots + b_n)$$

Q. Show that If $a \leq b$ and $x \leq y$ are real numbers then $2(ax + by) \geq (a + b)(x + y)$.

Ans: $a \leq b \Rightarrow a - b \leq 0$ and $x \leq y \Rightarrow x - y \leq 0$

$$\begin{aligned} &\Rightarrow (a - b)(x - y) \geq 0 \\ &\Rightarrow ax - ay - bx + by \geq 0 \\ &\Rightarrow ax + by \geq ay + bx \\ &\Rightarrow ax + by + ax + by \geq ay + bx + ax + by \\ &\Rightarrow 2(ax + by) \geq (a + b)(x + y) \text{ Proved} \end{aligned}$$

Q. Show that If $a \geq b$ and $x \leq y$ are real numbers then $2(ax + by) \leq (a + b)(x + y)$.

Ans: $a \geq b \Rightarrow a - b \geq 0$ and $x \leq y \Rightarrow x - y \leq 0$

$$\begin{aligned} &\Rightarrow (a - b)(x - y) \leq 0 \\ &\Rightarrow ax - ay - bx + by \leq 0 \\ &\Rightarrow ax + by \leq ay + bx \\ &\Rightarrow ax + by + ax + by \leq ay + bx + ax + by \\ &\Rightarrow 2(ax + by) \leq (a + b)(x + y) \text{ Proved} \end{aligned}$$

Q. Show that If $a \leq b \leq c$ and $x \leq y \leq z$ are real numbers then $3(ax + by + cz) \geq (a + b + c)(x + y + z)$.

Ans: $a \leq b \leq c \Rightarrow a - b \leq 0, b - c \leq 0, a - c \leq 0$ and $x \leq y \leq z \Rightarrow x - y \leq 0, y - z \leq 0, x - z \leq 0$

$$\begin{aligned} &\Rightarrow (a - b)(x - y) \geq 0 \\ &\Rightarrow ax - ay - bx + by \geq 0 \\ &\Rightarrow ax + by \geq ay + bx \text{ -----(1)} \end{aligned}$$

$$\text{Similarly } \Rightarrow by + cz \geq bz + cy \text{ -----(2) and } \Rightarrow az + cx \geq ax + cz$$

Adding (1), (2), (3) we get

$$\begin{aligned} &\Rightarrow ax + by + by + cz + ax + cz \geq ay + bx + by + cz + ax + cz \text{ and adding } ax + by + cz \text{ both side} \\ &\Rightarrow ax + by + by + cz + ax + cz + ax + by + cz \geq ay + bx + by + cz + ax + cz + ax + by + cz \\ &\Rightarrow 3(ax + by + cz) \geq (a + b + c)(x + y + z) \text{ (Proved).} \end{aligned}$$

Q. Show that If $a \geq b \geq c$ and $x \leq y \leq z$ are real numbers then $3(ax + by + cz) \leq (a + b + c)(x + y + z)$.

Ans: $a \geq b \geq c \Rightarrow a - b \geq 0, b - c \geq 0, a - c \geq 0$ and $x \leq y \leq z \Rightarrow x - y \leq 0, y - z \leq 0, x - z \leq 0$

$$\Rightarrow (a - b)(x - y) \leq 0$$

$$\Rightarrow ax - ay - bx + by \leq 0$$

$$\Rightarrow ax + by \leq ay + bx \text{ -----(1)}$$

$$\text{Similarly } \Rightarrow by + cz \leq bz + cy \text{ -----(2) and } \Rightarrow az + cx \leq ax + cz$$

Adding (1), (2), (3) we get

$$\Rightarrow ax + by + by + cz + ax + cz \leq ay + bx + by + cz + ax + cz \text{ and adding } ax + by + cz \text{ both side}$$

$$\Rightarrow ax + by + by + cz + ax + cz + ax + by + cz \leq ay + bx + by + cz + ax + cz + ax + by + cz$$

$$\Rightarrow 3(ax + by + cz) \leq (a + b + c)(x + y + z) \text{ (Proved).}$$

Exercise:

1. If $a, b, c > 0$, then prove that $a^3 + b^3 + c^3 \geq \frac{(a + b + c)^3}{9} \geq 3abc$.

2. If $a, b, c > 0$, then prove that $a^4 + b^4 + c^4 \geq abc(a + b + c)$.

3. If $a, b, c, d > 0$, then prove that $a^5 + b^5 + c^5 + d^5 \geq abcd(a + b + c + d)$.

4. Prove that $\sqrt{1} + \sqrt{2} + \sqrt{3} + \dots + \sqrt{n} \leq n\sqrt{\frac{n+1}{2}}$.

5. Prove that $\frac{1}{\sqrt{n}} \left(\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{n}} \right) \leq (2n-1)^{\frac{1}{4}}$.

6. If $a, b, c > 0$, then prove that $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}$.

7. Prove that $\left(\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} \right) \leq \sqrt{2n-1}$.

8. If $a, b, c > 0$, then prove that $(b+c)^3 + (c+a)^3 + (a+b)^3 \geq \frac{8(a+b+c)^3}{9}$.

9. If $a, b, c > 0$ such that $a + b + c = 4$, then prove that $a^3 + b^3 + c^3 \geq \frac{64}{9}$.

Functional Equation

Functional Equation: An equation which contains an unknown function is called as functional equation. Here we are going to solve some of the functional equation.

Basic Methods for Solving Functional Equation:

1. **Substituting the values for the variable.** Most common with 0 and 1. After that if possible make some part of the functional equation to become constant.
2. **Mathematical Induction:** In this method we can find $f(n)$ by using $f(1)$. After that find $f\left(\frac{1}{n}\right)$ and $f(r)$ for rational r . This is used when function is defined over rational numbers.
3. **By investing one-one and onto function in the equation:** In many cases it is not difficult to find this.
4. **By finding fixed point or zeroes of the function.**
5. By substituting the function.
6. By expressing function as sum of odd and even function.
7. By assuming solutions.
8. By recurrence relation.
9. By using Cauchy Equation:
 - (a) If $f(x+y) = f(x) + f(y)$, its solution is $f(x) = kx$.
 - (b) If $f(x+y) = f(x)f(y)$, its solution is $f(x) = a^x$.
 - (c) If $f(xy) = f(x) + f(y)$, its solution is $f(x) = \log x$
 - (d) If $f(xy) = f(x)f(y)$, its solution is $f(x) = x^n$
10. It is very important to guess the solution at the beginning. This will help a lot in finding the appropriate substitution. But don't forget to verify the solution.

Q. Solve $f(x+y) = f(x) + f(y)$ for all rational number x, y .

Ans: $f(x+y) = f(x) + f(y)$

By putting $x=0, y=0$ we get

$$\Rightarrow f(0) = f(0) + f(0) \Rightarrow f(0) = 0. \text{-----(1)}$$

Again by putting $y = -x$ we get

$$\Rightarrow f(0) = f(x) + f(-x) \Rightarrow 0 = f(x) + f(-x)$$

$$\Rightarrow f(-x) = -f(x) \text{-----(2)}$$

Let x is a positive integer, then

$$f(x) = f(1+1+\dots x \text{ times}) = f(1) + f(1) + \dots x \text{ times} = xf(1)$$

$$\Rightarrow f(x) = kx \text{-----(3) by taking } f(1) = k = \text{constant.}$$

Let x is a negative integer, then

$$f(x) = f(-n) = f(-1-1-\dots n \text{ times}) = f(-1) + f(-1) + \dots n \text{ times}$$

$$\Rightarrow f(x) = f(-n) = nf(-1) = -nf(1) = xf(1) = kx \text{ -----(4) by taking } f(1) = k = \text{constant.}$$

Let x is a rational number, then

$$f(p) = f\left(\frac{p}{q} + \frac{p}{q} + \dots q \text{ times}\right) = f\left(\frac{p}{q}\right) + f\left(\frac{p}{q}\right) + \dots q \text{ times} = qf\left(\frac{p}{q}\right).$$

$$\Rightarrow qf\left(\frac{p}{q}\right) = f(p) = kp$$

$$\Rightarrow f\left(\frac{p}{q}\right) = f(p) = k\left(\frac{p}{q}\right)$$

$$\Rightarrow f(x) = kx \text{ -----(5)}$$

Hence Solution $\Rightarrow f(x) = kx$ for all rational number x.

Q. Solve $f(x+y) = f(x)f(y)$ for all rational number x, y .

Ans: $f(x+y) = f(x)f(y)$

By putting $x=0, y=0$ we get

$$\Rightarrow f(0) = f(0)f(0) \Rightarrow (f(0))^2 - f(0) = 0 \Rightarrow f(0)(f(0)-1) = 0.$$

$$\Rightarrow f(0) = 0, 1 \text{ -----(1)}$$

First let us consider $f(0) = 0$

Then by putting $y=0$ in $f(x+y) = f(x)f(y)$ we get

$$\Rightarrow f(x+0) = f(x)f(0) \Rightarrow f(x) = f(x) \times 0 = 0$$

$$\Rightarrow f(x) = 0 \text{ -----(I)}$$

Now taking $f(0) = 1$

Again by putting $y = -x$ we get

$$\Rightarrow f(0) = f(x) \times f(-x) \Rightarrow 1 = f(x) \times f(-x)$$

$$\Rightarrow f(-x) = \frac{1}{f(x)} \text{ -----(2)}$$

Let x is a positive integer, then

$$f(x) = f(1+1+\dots x \text{ times}) = f(1) \times f(1) \times \dots x \text{ times} = (f(1))^x = k^x$$

$$\Rightarrow f(x) = k^x \text{ -----(3) by taking } f(1) = k = \text{constant.}$$

Let x is a negative integer, then

$$f(x) = f(-n) = f(-1-1-\dots n \text{ times}) = f(-1) \times f(-1) \times \dots n \text{ times}$$

$$\Rightarrow f(x) = f(-n) = (f(-1))^n = \left(\frac{1}{f(1)}\right)^n = (f(1))^{-n} = k^{-n} = k^x \text{ -----(4)}$$

by taking $f(1) = k = \text{constant.}$

Let x is a rational number, then

$$f(p) = f\left(\frac{p}{q} + \frac{p}{q} + \dots + \frac{p}{q} \text{ } q \text{ times}\right) = f\left(\frac{p}{q}\right) \times f\left(\frac{p}{q}\right) \times \dots \times f\left(\frac{p}{q}\right) \text{ } q \text{ times} = \left\{f\left(\frac{p}{q}\right)\right\}^q$$

$$\Rightarrow \left\{f\left(\frac{p}{q}\right)\right\}^q = k^p$$

$$\Rightarrow f\left(\frac{p}{q}\right) = (k^p)^{\frac{1}{q}} = k^{\frac{p}{q}}$$

$$\Rightarrow f(x) = k^x \text{ -----(5)}$$

Hence Solution $\Rightarrow f(x) = k^x$ for all rational number x .

So All solutions: $\Rightarrow f(x) = k^x$ and $\Rightarrow f(x) = 0$.

Exercise: Solve the following functional equation.

1. $f(1) = 2$ and $f(xy) = f(x)f(y) - f(x+y) + 1$.
2. $f(x+y) + f(x)f(y) = f(xy) + f(x) + f(y)$.
3. $xf(x) = f(f(x))$ and $f(f(x)) = 0$.
4. $f(1) = 2$, $f(2) = 4$ and $f(f(x) + f(y)) = f(f(x)) + f(y)$.
5. $f(xf(x) + f(y)) = x^2 + y$.
6. $f(x - f(y)) = f(f(y)) + xf(y) + f(x) - 1$.

The Principle of Inclusion - Exclusion

From the set we have

1. For two sets $|A \cup B| = |A| + |B| - |A \cap B|$

2. For three sets $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|$

3. For Four sets

$$|A \cup B \cup C \cup D| = |A| + |B| + |C| + |D| - |A \cap B| - |A \cap C| - |A \cap D| - |B \cap C| - |B \cap D| - |C \cap D| \\ + |A \cap B \cap C| + |A \cap B \cap D| + |A \cap C \cap D| + |B \cap C \cap D| - |A \cap B \cap C \cap D|$$

Similarly for n number of sets :

The Principle of Inclusion – Exclusion: - If $A_1, A_2, A_3, \dots, A_n$ are finite sets, then

$$|A_1 \cup A_2 \cup A_3 \cup \dots \cup A_n| = \sum_{1 \leq i \leq n} |A_i| - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| + \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k| - \dots \\ \dots + (-1)^{n+1} |A_1 \cap A_2 \cap A_3 \cap \dots \cap A_n|$$

Alternative form of Inclusion – Exclusion Principle: - If $A_1, A_2, A_3, \dots, A_n$ are subsets of finite sets S, then

$$\left| \bigcap_{i=1}^n A_i' \right| = |S| - \left| \bigcup_{i=1}^n A_i \right|$$

OR

$$\left| A_1' \cap A_2' \cap A_3' \cap \dots \cap A_n' \right| = |S| - \left(\sum_{1 \leq i \leq n} |A_i| - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| + \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k| - \dots \right. \\ \left. \dots + (-1)^{n+1} |A_1 \cap A_2 \cap A_3 \cap \dots \cap A_n| \right)$$

OR

$$N(A_1' A_2' A_3' \dots A_n') = N - \left(\sum_{1 \leq i \leq n} N(A_i) - \sum_{1 \leq i < j \leq n} N(A_i A_j) + \sum_{1 \leq i < j < k \leq n} N(A_i A_j A_k) - \dots \right. \\ \left. \dots + (-1)^{n+1} N(A_1 A_2 A_3 \dots A_n) \right)$$

Derangement: A derangement is a permutation of objects that leaves no object in its original position.

Theorem: - The number of derangement of n objects

$$= D_n = n! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \dots + (-1)^n \frac{1}{n!} \right)$$

Example: - How many positive integers less than 1000 exist which are divisible by 7 but not by 11?

Ans: - Numbers divisible by 7 but not by 11 = no. divisible by 7 – no. divisible by 77

$$= \left[\frac{1000}{7} \right] - \left[\frac{1000}{77} \right] \\ = 142 - 12 = 130 \text{ (Ans)}$$

Example: -50 cars are assembled in a factory with the following three options available: (1) air conditioner, (2) music system (3) television. It is known that 20 cars had air conditioners, 18 had music system and 15 had televisions. If 8 of those cars had all three options, find the smallest numbers of cars, which have none of these options.

Ans: - C: air conditioner, M: music system, T: television.

Minimum number of car have none of the three option

$$= 50 - \text{Max number of } |M \cup T \cup C| \quad \{\text{For max } M \cup T \cup C, \text{ we must have minimum of } M \cap T, T \cap C, C \cap M, \text{ but } M \cap T, T \cap C, C \cap M \geq 8.\}$$

So the minimum values of $M \cap T, T \cap C, C \cap M$ is 8.

and given that $M \cap T \cap C = 8$

$$\begin{aligned} &= 50 - (|M| + |T| + |C| - |M \cap T| - |T \cap C| - |C \cap M| + |M \cap T \cap C|) \\ &= 50 - (20 + 18 + 15 - 8 - 8 - 8 + 8) \\ &= 13. \text{ Ans.} \end{aligned}$$

Example: - Among 100 students, 22 study mathematics, 33 study physics, 44 study biology, 11 study mathematics and biology, 7 study mathematics and physics, 15 study physics and biology: 28 do not study either.

(i) Find the number of students studying all three subjects.

(ii) Find the number of students studying exactly one of the three subjects.

Ans: - Given $|E| = 100, |M| = 22, |P| = 33, |B| = 44, |M \cap B| = 11, |M \cap P| = 7, |P \cap B| = 15,$

$$|M \cup P \cup B|' = 28.$$

$$\Rightarrow |M \cup P \cup B| = E - |M \cup P \cup B|' = 100 - 28 = 72$$

$$\Rightarrow |M| + |P| + |B| - |M \cap B| - |M \cap P| - |P \cap B| + |M \cap P \cap B| = 72$$

$$\Rightarrow 22 + 33 + 44 - 11 - 7 - 15 + |M \cap P \cap B| = 72$$

$$\Rightarrow |M \cap P \cap B| = 72 - 66 = 6$$

So, number of students studying all three subjects = 6.

Number of students studying exactly one of the three subjects

$$= |M \cup P \cup B| - |M \cap P| - |M \cap B| - |P \cap B| - 2|M \cap P \cap B|$$

$$= 72 - 7 - 11 - 15 - 2 \times 6$$

$$= 24 \text{ (Ans)}$$

Example: - Among 100 students, 32 study Mathematics, 20 study physics, 45 study Biology, 15 study Mathematics and Biology, 7 study Mathematics and Physics, 10 study Physics and Biology and 30 do not study any of the three subjects.

(i) Find the number of students studying all three subjects.

(ii) Find the number of students studying exactly one of the three subjects.

Ans:- Given $|E| = 100, |M| = 32, |P| = 20, |B| = 45, |M \cap B| = 15, |M \cap P| = 7, |P \cap B| = 10,$

$$|M \cup P \cup B|' = 30.$$

$$\Rightarrow |M \cup P \cup B| = E - |M \cup P \cup B|' = 100 - 30 = 70$$

$$\Rightarrow |M| + |P| + |B| - |M \cap B| - |M \cap P| - |P \cap B| + |M \cap P \cap B| = 70$$

$$\Rightarrow 32 + 20 + 45 - 15 - 7 - 10 + |M \cap P \cap B| = 70 \Rightarrow |M \cap P \cap B| = 70 - 65 = 5$$

So, number of students studying all three subjects = 5.

Number of students studying exactly one of the three subjects

$$= |M \cup P \cup B| - |M \cap P| - |M \cap B| - |P \cap B| - 2|M \cap P \cap B|$$

$$= 70 - 7 - 15 - 10 - 2 \times 5 = 28 \quad (\text{Ans})$$

Example: - Explain the principle of inclusion exclusion. Using this principle, find out the number of solutions for the equation $x_1 + x_2 + x_3 = 13$, where x_1, x_2, x_3 are non-negative integers less than six.

Ans: -The principle of inclusion-exclusion: -

$$|A_1 \cup A_2 \cup \dots \cup A_n| = \sum_{i \leq n} |A_i| - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| + \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k| - \dots$$

$$\dots + (-1)^n |A_1 \cap A_2 \cap \dots \cap A_n|$$

$$x_1 + x_2 + x_3 = 13 \text{ and } x_1 < 6, x_2 < 6, x_3 < 6.$$

$$\text{Let } P_1 : x_1 \geq 6, P_2 : x_2 \geq 6, P_3 : x_3 \geq 6,$$

Now Solution for $x_1 < 6, x_2 < 6, x_3 < 6$ is

$$N(P_1'P_2'P_3') = N - N(P_1) - N(P_2) - N(P_3) + N(P_1P_2) + N(P_2P_3) + N(P_3P_1) - N(P_1P_2P_3)$$

$$= C(15,13) - C(9,7) - C(9,7) - C(9,7) + C(3,1) + C(3,1) + C(3,1) - 0$$

$$= C(15,13) - 3C(9,7) + 3C(3,1)$$

$$= 105 - 3 \times 36 + 3 \times 3 = 105 - 108 + 9 = 6 \quad (\text{Ans})$$

Example: - Find the number of integer solutions of the equation $x_1 + x_2 + x_3 + x_4 = 17$, where each $x_i \geq 0$ and $x_1 \leq 3, x_2 \leq 4, x_3 \leq 5$ and $x_4 \leq 8$.

Ans: -

$$\text{Let } P_1 : x_1 > 3$$

$$P_2 : x_2 > 4$$

$$P_3 : x_3 > 5$$

$$P_4 : x_4 > 8$$

No. of satisfying the inequality $x_1 \leq 3, x_2 \leq 4, x_3 \leq 5$ and $x_4 \leq 8$ is

$$N(P_1'P_2'P_3'P_4') = N - N(P_1) - N(P_2) - N(P_3) - N(P_4) + N(P_1P_2) + N(P_1P_3) +$$

$$N(P_1P_4) + N(P_2P_3) + N(P_2P_4) + N(P_3P_4) - N(P_1P_2P_3) - N(P_1P_2P_4) - N(P_1P_3P_4)$$

$$- N(P_2P_3P_4) + N(P_1P_2P_3P_4)$$

$$= C(4+17-1,17) - C(4+13-1,13) - C(4+12-1,12) - C(4+11-1,11) - C(4+8-1,8)$$

$$+ C(4+8-1,8) + C(4+7-1,7) + C(4+4-1,4) + C(4+6-1,6) + C(4+3-1,3) + C(4+2-1,2)$$

$$- C(4+2-1,2) - 0 - 0 - 0 + 0$$

$$= C(20,17) - C(16,13) - C(15,12) - C(14,11) - C(11,8) + C(11,8) + C(10,7)$$

$$+ C(7,4) + C(9,6) + C(6,3) + C(5,2) - C(5,2)$$

$$= 1140 - (560 + 455 + 363 + 165) + (165 + 120 + 35 + 84 + 20 + 10) - 10$$

$$= 20 \quad (\text{Ans})$$

Example: - Explain the principle of inclusion exclusion. Using this principle, find out the number of solutions for the equation $x_1 + x_2 + x_3 = 13$, where x_1, x_2, x_3 are non-negative integers less than six.

Ans: -The principle of inclusion-exclusion: -

$$|A_1 \cup A_2 \cup \dots \cup A_n| = \sum_{i \leq n} |A_i| - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| + \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k| - \dots \dots \dots + (-1)^n |A_1 \cap A_2 \cap \dots \cap A_n|$$

$$x_1 + x_2 + x_3 = 13 \text{ and } x_1 < 6, x_2 < 6, x_3 < 6.$$

$$\text{Let } P_1 : x_1 \geq 6, P_2 : x_2 \geq 6, P_3 : x_3 \geq 6,$$

Now Solution for $x_1 < 6, x_2 < 6, x_3 < 6$ is

$$\begin{aligned} N(P_1'P_2'P_3') &= N - N(P_1) - N(P_2) - N(P_3) + N(P_1P_2) + N(P_2P_3) + N(P_3P_1) - N(P_1P_2P_3) \\ &= C(15,13) - C(9,7) - C(9,7) - C(9,7) + C(3,1) + C(3,1) + C(3,1) - 0 \\ &= C(15,13) - 3C(9,7) + 3C(3,1) \\ &= 105 - 3 \times 36 + 3 \times 3 = 105 - 108 + 9 = 6 \text{ (Ans)} \end{aligned}$$

Example: - Find the number of integer solutions of the equation $x_1 + x_2 + x_3 + x_4 = 17$, where each $x_i \geq 0$ and $x_1 \leq 3, x_2 \leq 4, x_3 \leq 5$ and $x_4 \leq 8$.

Ans: -

$$\text{Let } P_1 : x_1 > 3$$

$$P_2 : x_2 > 4$$

$$P_3 : x_3 > 5$$

$$P_4 : x_4 > 8$$

No. of satisfying the inequality $x_1 \leq 3, x_2 \leq 4, x_3 \leq 5$ and $x_4 \leq 8$ is

$$\begin{aligned} N(P_1'P_2'P_3'P_4') &= N - N(P_1) - N(P_2) - N(P_3) - N(P_4) + N(P_1P_2) + N(P_1P_3) + \\ &N(P_1P_4) + N(P_2P_3) + N(P_2P_4) + N(P_3P_4) - N(P_1P_2P_3) - N(P_1P_2P_4) - N(P_1P_3P_4) \\ &- N(P_2P_3P_4) + N(P_1P_2P_3P_4) \\ &= C(4+17-1,17) - C(4+13-1,13) - C(4+12-1,12) - C(4+11-1,11) - C(4+8-1,8) \\ &+ C(4+8-1,8) + C(4+7-1,7) + C(4+4-1,4) + C(4+6-1,6) + C(4+3-1,3) + C(4+2-1,2) \\ &- C(4+2-1,2) - 0 - 0 - 0 + 0 \\ &= C(20,17) - C(16,13) - C(15,12) - C(14,11) - C(11,8) + C(11,8) + C(10,7) \\ &+ C(7,4) + C(9,6) + C(6,3) + C(5,2) - C(5,2) \\ &= 1140 - (560 + 455 + 363 + 165) + (165 + 120 + 35 + 84 + 20 + 10) - 10 \\ &= 20 \text{ (Ans)} \end{aligned}$$

Example: - Find how many derangements of 5 elements?

Ans: - The number of derangement =

$$D_5 = 5! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} \right) = 120 - 120 + 60 - 20 + 5 - 1 = 44$$

Example: - Suppose 7 people randomly arrive at a dark room to retrieve their hats. Find the number of ways such that no person picks his own hat.

Ans: - Number of ways

$$= D_7 = 7! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \frac{1}{6!} - \frac{1}{7!} \right) = 0 + 2520 - 840 + 210 - 42 + 7 - 1 = 1854 \text{ (ans).}$$

Exercise

1. A discrete mathematics class contains 25 students majoring in computer science, 13 students majoring in mathematics and 8 joint mathematics and computer science majors, how many students are in this class?
2. Let that there are 1807 freshmen at your school. Out of these 453 are taking course in computer science, 567 are taking course in mathematics and 299 are taking in both computer science and mathematics. How many are not taking a course either in mathematics or in computer science?
3. A total of 1232 students have taken a course in Spanish, 879 have taken in French, 114 have taken in Russian, 103 have taken in both Spanish and French, 23 have taken in both Spanish and Russian and 14 have taken in both French and Russian. If 2902 students have taken at least one of Spanish, French and Russian, how many students have taken in all three languages?
4. There are 345 students at a college who have taken a course in calculus, 212 have taken in discrete mathematics and 188 have taken in both calculus and discrete mathematics. How many students have taken course in either calculus or discrete mathematics?
5. Find $|A \cup B| = ?$ if $|A| = 12$, $|B| = 18$.
 - (i) $|A \cap B| = 1$
 - (ii) $|A \cap B| = 6$
 - (iii) $A \cap B = \phi$
 - (iv) $A \subseteq B$
6. How many positive integers not exceeding 1000 are divisible by 7 or 11?
7. How many positive integers not exceeding 100 that are not divisible 5 or by 7?
8. Find the number of positive integers not exceeding 100 that are either odd or the square of integers.
9. Find the number of positive integers not exceeding 1000 that are either the square or the cube of an integers.
10. A survey of households in the United States reveals that 96% have atleast one television set, 98% have telephone service and 95% have telephone service and atleast one television set, what % of households in the United states have neither telephone nor a television set?
11. A marketing report concerning personal computers states that 650000 owners will buy a modem for their machine s next year and 1250000 will buy atleast one software package. If the report states that 1450000 owners will buy either a modem or atleast one software package, how many will buy both a modem and atleast one software package?
12. Write the explicit formula for principle of inclusion exclusion for 5 sets.
13. How many elements are in the union of four sets if each set contains 50, 60, 70, 80 elements respectively, each pair of sets has 5 elements in common, each triple of sets has 1 common element and no element common in all four sets?
14. How many terms are there in the formula for the number of elements in the union of ten sets?
15. In a survey of 270 college students, it is found that 64 like Brussels sprouts, 94 likes broccoli, 58 likes cauliflower, 26 like both Brussels sprout and broccoli, 28 likes Brussels sprout and cauliflower,

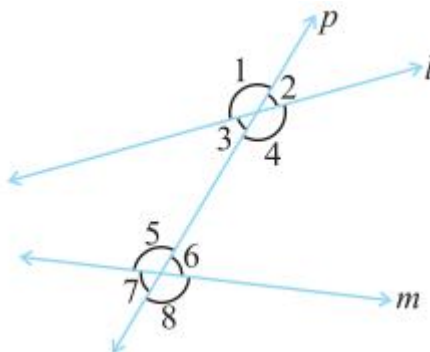
- 22 like both broccoli and cauliflower and 14 likes all three vegetables. How many of the 270 students do not like any of these vegetables?
16. At an art academy there are 45 students taking ceramics, 55 students taking painting and 30 students taking sculpture. There are 15 students in both ceramic and painting, 10 in both painting and sculpture, 5 in both ceramic and sculpture and 12 taking all the three courses. How many students are taking atleast one course at the art academy?
17. Find the number of integers between 1 and 250 that are divisible by any of the integers 2, 3, 5 and 7.
18. How many numbers from 1 to 1000 are not divisible by 2, 3 and 5?
19. How many numbers from 1 to 1000 are not divisible by 4, 5 and 6?
20. Among integers 1 to 300, how many of them are not divisible by 3, nor by 5, nor by 7? How many of them are divisible by 3, but not by 5 nor by 7?
21. At a meeting of 30 women, 17 are descended from George Washington, 16 are descended from John Adams and 5 are not descended from Washington or Adams. How many of the 30 women are descended from both Washington and Adams?
22. 75 children went to an amusement park where they can ride on the merry-go-round, roller coaster and ferris wheel. It is known 20 of them have taken all three rides, 55 of them taken atleast two of three rides. Each rides costs \$0.50 and the total receipt of the amusement park was \$70. Determine the number of children who did not try any of the rides.
23. A survey was conducted among 1000 people, out of which 595 are democrats, 595 wear glasses and 550 like ice-cream, 395 of them are democrats who wear glasses, 350 of them are democrats who like ice-cream and 400 of them wear glasses and like ice-cream, 250 of them are democrats who wear glasses and also like ice-cream. How many of them who are not democrats, do not wear glasses and do not like ice-cream? How many of them are democrats who do not wear glasses and do not like ice-cream?
24. 30 cars were assembled in a factory. The options available were a radio, an air conditioner and white-wall tires. It is known that 15 of the cars have radio, 8 of them have air conditioner and 6 of them have white-wall tires. Moreover, 3 of them have all three options. Then find out atleast how many cars do not have any options?
25. Out of 200 students 50 Of them take Discrete, 140 of them take Economics and 24 of them take both courses. Since both courses have scheduled examinations for the following day, only one student are not in either of these courses will be able to go to part the night before. How many students will be at the party?
26. Find the number of integer solutions of the equation $x_1 + x_2 + x_3 = 13$, where each $x_i \geq 0$ and $x_1 \leq 4, x_2 \leq 5, x_3 \leq 6$.
27. How many integer solution are there to the equation $x_1 + x_2 + x_3 + x_4 = 13$, where each $0 \leq x_i \leq 5$.
28. How many onto functions are there from a set with seven elements to one with five elements?
29. How many ways 6 toys can be distributed among 3 children such each child gets atleast one toy?
30. In how many ways 8 balls can be distributed in 3 distinct urns if each urn contains atleast one ball?
31. How many ways are there to assign six different jobs to five different employees if every employee is assigned at least one job?
32. List all derangement of 1234.
33. How many derangements are there for 7 elements?

34. In how many ways none of the 10 people receive the correct hat if someone returns their hat randomly?
35. Find the number of ways of putting 5 letters in five addressed envelopes such that no letter is placed in the right envelope?
36. In how many ways can the four walls of a room be painted with three colours so that no two adjacent walls have the same colour?
37. In how many number of ways can three examinations be scheduled within five days period with no restriction on the number of examinations scheduled each day?
38. Suppose 8 persons are given greeting cards. These cards are collected, shuffled again distributed to the persons, one to each. Find the number of ways in which no persons gets the same card again.
39. Let 6 peoples randomly arrive at a dark room to retrieve their hats. Find the number of ways such that exactly two persons pick up their own hats.
40. How many solutions does the equation $x_1 + x_2 + x_3 = 11$ have, where x_1, x_2, x_3 are non-negative integers?

Geometry

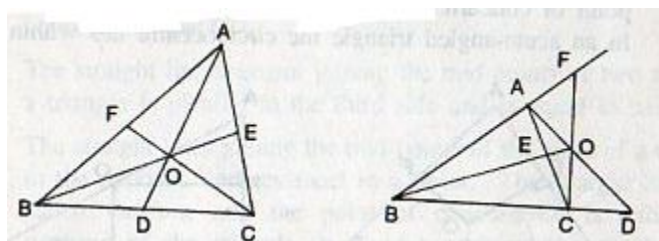
1. Angles:

- (a) Two rays OA and OB with common point O form an angle AOB denoted as $\angle AOB$.
- (b) Angle can be measured by degrees, minute and second. $1^\circ = 60'$, $1' = 60''$.
- (c) Right Angle: An angle whose measurement is 90° is called as right angle.
- (d) Acute angle: An angle whose measurement is greater than 0° and less than 90° is called as acute angle.
- (e) Obtuse angle: An angle whose measurement is greater than 90° and less than 180° is called as obtuse angle.
- (f) Straight angle: An angle whose measurement is 180° is called as straight angle.
- (g) Reflex angle: An angle whose measurement is greater than 180° and less than 360° is called as reflex angle.
- (h) Complete angle: An angle whose measurement is 360° is called as complete angle.
- (i) Equal Angles: Two angles are said to be equal if they have equal measurement.
- (j) Complementary angles: Two angles are said to be complementary if their sum is 90° .
- (k) Supplementary angles: Two angles are said to be supplementary if their sum is 180° .
- (l) Adjacent angles: Two angles are said to be adjacent if they have one common arm and one common vertex.
- (m) If a ray stands on a straight line, then sum of two adjacent angles is 180° .
- (n) The sum of all angles around a point is 360° .
- (o) **Vertically opposite angles:** If two lines intersect at a point, then two pairs of vertically opposite angles so formed are equal to each other.
- (p) **Parallel Lines:** If two lines on the same plane do not intersect when produced either side are called as parallel lines.
- (q) **Angles Formed when a transversal line intersect two lines:**



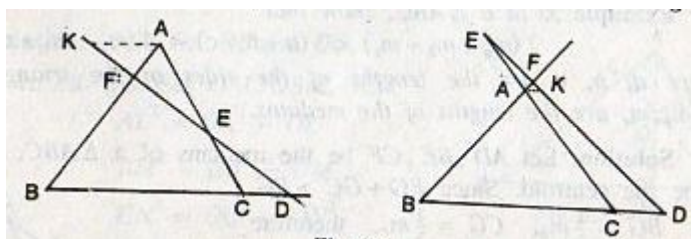
Interior angles	$\angle 3, \angle 4, \angle 5, \angle 6$
Exterior angles	$\angle 1, \angle 2, \angle 7, \angle 8$
Pairs of Corresponding angles	$\angle 1$ and $\angle 5$, $\angle 2$ and $\angle 6$, $\angle 3$ and $\angle 7$, $\angle 4$ and $\angle 8$
Pairs of Alternate interior angles	$\angle 3$ and $\angle 6$, $\angle 4$ and $\angle 5$
Pairs of Alternate exterior angles	$\angle 1$ and $\angle 8$, $\angle 2$ and $\angle 7$
Pairs of interior angles on the same side of the transversal	$\angle 3$ and $\angle 5$, $\angle 4$ and $\angle 6$

- (r) **When a transversal line intersect two parallel lines:**
Pairs of Alternate Interior angles are equal.
Pairs of Alternate Exterior angles are equal.
Pairs of corresponding angles are equal.
Pairs of interior angles on the same side of the transversal are supplementary.
Pairs of exterior angles on the same side of the transversal are supplementary.
2. **Incentre:** The point of concurrency of angle bisectors of a triangle is called as Incentre. Incentre of the triangle always lies within the triangle.
 3. **Circumcentre:** The point of concurrency of perpendicular bisectors sides of a triangle is called as Circumcentre.
In acute angled triangle the circumcentre lies within the triangle.
In a right triangle the Circumcentre lies on the hypotenuse of the right triangle.
In an obtuse angled triangle circumcentre lies outside the triangle.
 4. **Orthocentre:** The point of concurrency of altitudes of a triangle is called as Orthocentre.
In acute angled triangle the orthocentre lies within the triangle.
In a right triangle the orthocentre is the vertex opposite to the hypotenuse of the right triangle.
In an obtuse angled triangle orthocentre lies outside the triangle.
 5. **Centroid:** The point of concurrency of medians of a triangle is called as Centroid.
Centroid of the triangle always lies inside the triangle.
Centroid of the triangle divides the median in the ratio 2:1.
 6. **Congruence Triangles:** Two triangles are congruent if they have same shape and size. For congruent triangles their corresponding sides and angles are equal.
 7. **SSS Congruent Condition:** Two triangles are congruent if three sides of one triangle are respectively equal to the three sides of the other triangle.
 8. **SAS Congruent Condition:** Two triangles are congruent if two sides and included angle of one triangle are respectively equal to the two sides and included angle of the other triangle.
 9. **ASA Congruent Condition:** Two triangles are congruent if two angles and included side of one triangle are respectively equal to the two angles and included side of the other triangle.
 10. **RHS Congruent Condition:** Two right triangles are congruent if the hypotenuse and one of the side of one triangle are respectively equal to the three hypotenuse and one of the side of the other triangle.
 11. **Similar Triangles:** Two triangles are said to be similar if they have same shape but sizes are different. If two triangles are similar then, their corresponding angles are equal and ratio of their corresponding sides are equal.
 12. **AA Similarity Criteria:** If two angles of one triangle are equal to the corresponding two angles of another triangle, then that two triangles are similar.
 13. **SSS Similarity Criteria:** If all the corresponding sides of two triangles are equal in ratio, then two triangles are similar.
 14. **SAS Similarity Criteria:** Two sides have lengths in the same ratio, and the angles included between these sides have the same measure, then that two triangles are similar to each other.
 15. **Cevas' Theorem:** If three concurrent straight lines are drawn from the angular point of a triangle to meet the opposite sides, then the product of three alternate segments taken in order is equal to the product of the other three segments.



$$BD.CE.AF = DC.EA.FB$$

16. **Menelaus' Theorem:** If a transversal drawn to cut the sides, or sides produced, of a triangle, the product of three alternate segments taken in order is equal to the product of the other three segments.



$$AF.BD.CE = EA.FB.DC$$

17. **Properties of Triangle:**

- (a) Sum of all the interior angles of a triangle is 180° .
- (b) The longest side of a triangle is opposite to the greatest angle and vice versa. The least side of triangle is opposite to the smaller angle and vice versa.
- (c) Sum of any two sides of a triangle is always greater than the third side.
- (d) If one angle of a triangle is greater than the second angle then the side opposite to the first is greater than the side opposite to second.
- (e) Exterior angle of a triangle is equal to the sum of two interior opposite angles.
- (f) A scalene triangle is one which has all sides are unequal.
- (g) An isosceles triangle has two of its sides are equal.
- (h) An equilateral triangle has all the three sides are equal.
- (i) An obtuse angled triangle has one of its angle is obtuse.
- (j) An acute angled triangle has all of its angles are acute.
- (k) A right angled triangle has one right angle.
- (l) There can be only one right angle in a triangle.
- (m) There can be only one obtuse angle in a triangle.
- (n) Angles opposite to equal sides of a triangle are equal.
- (o) Sides opposite to equal angles of a triangle are equal.
- (p) The line segment joining the mid points of two sides of a triangle is parallel to third and half of third side.
- (q) A line drawn through the midpoint of two sides of a triangle is parallel to the third side and half of third side.
- (r) In a triangle ABC, if $BC^2 = AB^2 + AC^2$, then $\angle A$ is right angle.
- (s) In a triangle ABC, if $BC^2 < AB^2 + AC^2$, then $\angle A$ is acute angle.
- (t) In a triangle ABC, if $BC^2 > AB^2 + AC^2$, then $\angle A$ is obtuse angle.
- (u) Triangles on the same base and same parallels are equal in area.

- (v) If two triangles, have a common vertex and their bases are along the same straight line, then ratio of their areas is equal to the ratio of their base.
- (w) In any triangle the sum of squares on two sides of a triangle is equal to the twice the sum of the squares on half the third side and median to the third side.

18. Properties of Parallelogram:

- (a) Opposite sides of a parallelogram are equal.
- (b) Opposite angles of a parallelogram are equal.
- (c) A pair of adjacent angles of a parallelogram is supplementary.
- (d) A diagonal divided the parallelogram into two congruent triangles.
- (e) Diagonals of a parallelogram bisect each other.
- (f) If a parallelogram can be inscribed in a circle, then it is a rectangle.
- (g) If two parallelograms are on same base and same parallels then they are equal in area.

19. Properties of Rectangle:

- (a) Opposite sides of a rectangle are equal.
- (b) Each angles of a rectangle is right angle.
- (c) Diagonals of a rectangle bisect each other.
- (d) Diagonals of a rectangle are equal.

20. Properties of a square:

- (a) All the angles of a square are right angles.
- (b) All the sides of a square are equal.
- (c) Diagonals of a square are equal.
- (d) Diagonal bisect each other.
- (e) Diagonals of a square are perpendicular to each other.

21. Properties of Rhombus:

- (a) All the sides of the rhombus are equal.
- (b) Opposite angles of the rhombus are equal.
- (c) Diagonal of a rhombus bisect each other.
- (d) Diagonals of a rhombus are perpendicular to each other.
- (e) Diagonal bisects the angles at the vertices.
- (f) If a rhombus can be inscribed in a circle, then it is a square.

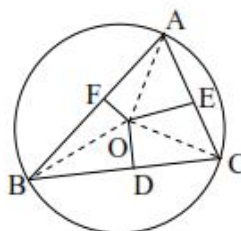
22. Properties of Parallel Lines:

- (a) Two lines in a plane which are parallel to the given line is parallel to each other.
- (b) Two intersecting lines cannot both parallel to the same line.
- (c) Two lines in the same plane, which are perpendicular to a given line in the same plane, are parallel to each other.
- (d) Three or more parallel lines makes equal intercepts on any transversal then they make equal intercepts on any other transversal also.

- (e) If a line parallel to one of the side of a triangle intersects other two sides in the two different points then intercepts proportional to these two sides.
- (f) If three or more parallel lines intersects two transversal, then the intercepts made on one transversal are proportional to the intercepts made on other transversal.

23. Properties of a Circle;

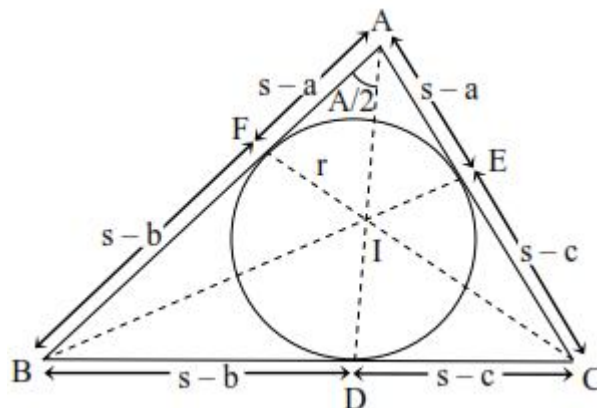
- (a) One and only one circle can pass through three non-collinear points.
- (b) Perpendicular from the centre of a circle to any chord, bisect the chord.
- (c) The line segment joining the centre of a circle to the midpoint of the chord is perpendicular to the chord.
- (d) Equal chords of a circle subtend equal angles at the centre.
- (e) In a circle, chords which subtend equal angles at centre are equal.
- (f) Equal chords of a circle are equidistant from the centre.
- (g) Chords which are equidistant from centre are equal.
- (h) An angle subtended by an arc at the centre is twice the angle subtended by it at any point on the remaining part of the circle.
- (i) If two chords subtend equal angles at the circumference of a circle, the chords are equal.
- (j) Angles in the same segment of a circle are equal.
- (k) Angle on a semicircle is a right angle.
- (l) Angle in a segment smaller than a semi circle is obtuse.
- (m) Angle in a segment greater than a semi circle is acute.
- (n) The line perpendicular to the tangent at the point of contact passes through the centre of the circle.
- (o) The line drawn perpendicular to the radius through the end point where the radius meets the circle is tangent to the circle at the end point.
- (p) The lengths of the two tangents drawn from an external point to the circle are equal.
- (q) If two chords of a circle intersect, whether inside or outside the circle, the rectangles contained by the segments of chords are equal in area.
- (r) If through a point on the circle, a tangents and a chord is drawn, then the angle which the tangent makes with the chord is equal to the angle in the alternate segments.
- (s) If from an external point P to a circle PAB & PCD are secant lines then $PA \times PB = PC \times PD$.
- (t) If from an external point P to a circle PAB is a secant line and PT is a tangent to the circle then $PA \times PB = PT^2$.
- (u) **Circum-circle of a triangle and its radius:**



$$(1) \quad R = \frac{a}{2 \sin A} = \frac{b}{2 \sin B} = \frac{c}{2 \sin C}.$$

$$(2) \quad R = \frac{abc}{4\Delta}, \text{ where } \Delta \text{ is the area of triangle.}$$

(v) **In-circle of a triangle and its radius:**



$$(1) \quad r = \frac{\Delta}{s}, \text{ where } \Delta \text{ is the area of triangle and } s \text{ is the perimeter of triangle.}$$

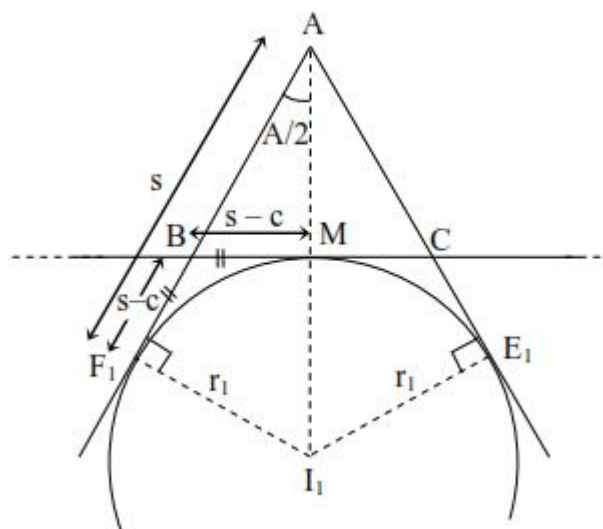
$$(2) \quad r = 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}.$$

$$(3) \quad r = (s - a) \tan \frac{A}{2} = (s - b) \tan \frac{B}{2} = (s - c) \tan \frac{C}{2}.$$

$$(4) \quad r = \frac{a \sin \frac{B}{2} \sin \frac{C}{2}}{\cos \frac{A}{2}} = \frac{b \sin \frac{C}{2} \sin \frac{A}{2}}{\cos \frac{B}{2}} = \frac{c \sin \frac{A}{2} \sin \frac{B}{2}}{\cos \frac{C}{2}}.$$

$$(5) \quad \cos A + \cos B + \cos C = 1 + \frac{r}{R}.$$

(w) **Ex-circle of a triangle and its radius:**



$$(1) \quad r_1 = \frac{\Delta}{s - a} = \frac{a \cos \frac{B}{2} \cos \frac{C}{2}}{\cos \frac{A}{2}} = s \tan \frac{A}{2} = (s - c) \cot \frac{B}{2} = (s - b) \cot \frac{C}{2} = \frac{a}{\tan \frac{B}{2} + \tan \frac{C}{2}}.$$

$$(2) \quad r_2 = \frac{\Delta}{s-b} = \frac{b \cos \frac{C}{2} \cos \frac{A}{2}}{\cos \frac{B}{2}} = s \tan \frac{B}{2} = (s-c) \cot \frac{A}{2} = (s-a) \cot \frac{C}{2} = \frac{b}{\tan \frac{C}{2} + \tan \frac{A}{2}}.$$

$$(3) \quad r_3 = \frac{\Delta}{s-c} = \frac{c \cos \frac{A}{2} \cos \frac{B}{2}}{\cos \frac{C}{2}} = s \tan \frac{C}{2} = (s-a) \cot \frac{B}{2} = (s-b) \cot \frac{A}{2} = \frac{c}{\tan \frac{A}{2} + \tan \frac{B}{2}}.$$

$$(4) \quad r_1 + r_2 + r_3 - r = 4R.$$

$$(5) \quad \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} = \frac{1}{r}.$$

$$(6) \quad \frac{1}{r^2} + \frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2} = \frac{a^2 + b^2 + c^2}{\Delta^2}.$$

$$(7) \quad \frac{1}{bc} + \frac{1}{ca} + \frac{1}{ab} = \frac{1}{2Rr}.$$

$$(8) \quad r_1 r_2 + r_2 r_3 + r_3 r_1 = s^2.$$

$$(9) \quad \Delta = 2R^2 \sin A \sin B \sin C.$$

$$(10) \quad \Delta = 4Rr \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}.$$

$$(11) \quad r_1 = 4R \sin \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}.$$

$$(12) \quad r_2 = 4R \cos \frac{A}{2} \sin \frac{B}{2} \cos \frac{C}{2}.$$

$$(13) \quad r_3 = 4R \cos \frac{A}{2} \cos \frac{B}{2} \sin \frac{C}{2}.$$

(x) **Distance of Circumcentre (O) from the Orthocentre(H), Incentre (I) and Excentre I_1, I_2, I_3 :**

(1) **Distance from Circumcentre (O) and Orthocentre (H):**

$$OH = R\sqrt{1-8\cos A \cos B \cos C}.$$

(2) **Distance from Circumcentre (O) and Incentre (I):**

$$OI = R\sqrt{1-8\sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}} = \sqrt{R}\sqrt{R-2r}$$

(3) **Distance from Circumcentre (O) and Excentre (I_1):**

$$OI_1 = R\sqrt{1+8\sin \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}} = \sqrt{R}\sqrt{R+2r_1}$$

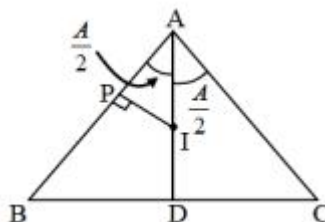
(4) **Distance from Circumcentre (O) and Excentre (I_2):**

$$OI_2 = R\sqrt{1+8\cos \frac{A}{2} \sin \frac{B}{2} \cos \frac{C}{2}} = \sqrt{R}\sqrt{R+2r_2}$$

(5) **Distance from Circumcentre (O) and Excentre (I_3):**

$$OI_1 = R\sqrt{1 + 8\cos\frac{A}{2}\cos\frac{B}{2}\sin\frac{C}{2}} = \sqrt{R}\sqrt{R + 2r_3}$$

(y) **Distance of Incentre from the vertices of the Triangle:**



Let I be the incentre. Let $IP \perp AB$. Hence $IP = r \angle PAI = \frac{A}{2}$.

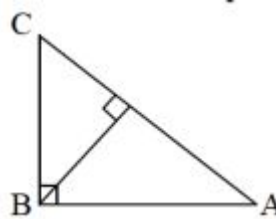
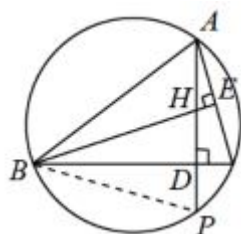
From right triangle IPA, $\sin \frac{A}{2} = \frac{r}{AI} \Rightarrow AI = r \operatorname{cosec} \frac{A}{2}$

$$(1) AI = r \operatorname{cosec} \frac{A}{2}$$

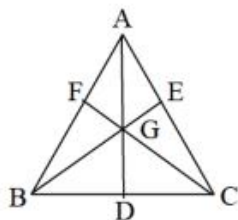
$$(2) BI = r \operatorname{cosec} \frac{B}{2}$$

$$(3) CI = r \operatorname{cosec} \frac{C}{2}$$

(z) **Note Points:**



- (1) The centroid of any triangle divides the join of circumcentre and orthocentre internally in the ratio 1:2.
- (2) If H is the orthocentre of triangle ABC and AH produced meets BC at D and the circumcentre of triangle ABC at P, then $HD = DP$.
As $BD = c \cos B$ and $DP = BD \cot C$
So, $DP = c \cos B \cot C = 2R \cos B \cos C$.
- (3) The orthocentre of an acute angled triangle is the incentre of the pedal triangle.
- (4) The centre of circumcircle falls inside the triangle if triangle is acute angled but outside when it is obtuse angled. If the triangle is right angled triangle the centre lies on midpoint of the hypotenuse.
- (5) The orthocentre falls inside the triangle if triangle is acute angled and outside if triangle is obtuse angled triangle. If the triangle is right angled then orthocentre (B) lies on the triangle.



(6) The length of median AD, BE and CF of triangle are given by:

(a) $AD = \frac{1}{2} \sqrt{2b^2 + 2c^2 - a^2} = \frac{1}{2} \sqrt{b^2 + c^2 + 2bc \cos A}.$

(b) $BE = \frac{1}{2} \sqrt{2c^2 + 2a^2 - b^2} = \frac{1}{2} \sqrt{c^2 + a^2 + 2ca \cos B}.$

(c) $CF = \frac{1}{2} \sqrt{2a^2 + 2b^2 - c^2} = \frac{1}{2} \sqrt{a^2 + b^2 + 2ab \cos C}.$

(d) Distance between circum centre (O) and in-centre (I) of triangle ABC is

$$OI = R \sqrt{1 - 8 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}}.$$

(e) Distance between circum centre (O) and ex-centre (I_1) opposite to angle A of

triangle ABC is $OI_1 = R \sqrt{1 + 8 \sin \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}}.$

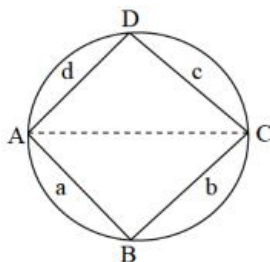
(f) Distance between circum centre (O) and ex-centre (I_2) opposite to angle B of

triangle ABC is $OI_2 = R \sqrt{1 + 8 \cos \frac{A}{2} \sin \frac{B}{2} \cos \frac{C}{2}}.$

(g) Distance between circum centre (O) and ex-centre (I_3) opposite to angle C of

triangle ABC is $OI_3 = R \sqrt{1 + 8 \cos \frac{A}{2} \cos \frac{B}{2} \sin \frac{C}{2}}.$

- (aa) **Cyclic Quadrilateral:** If all the four vertices of a quadrilateral lies on the circumference of the circle, then the quadrilateral is called as cyclic quadrilateral. OR A quadrilateral ABCD is said to be cyclic quadrilateral if there exists a circle passing through all its four vertices A, B, C and D. Let a cyclic quadrilateral with $Ab = a$, $BC = b$, $CD = c$, $DA = d$. Let $2s = a + b + c + d$.

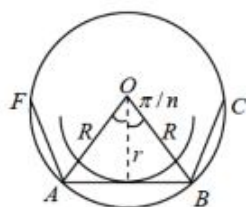


(1) $\angle B + \angle D = 180^\circ.$

(2) $\angle A + \angle C = 180^\circ.$

- (3) Area of cyclic quadrilateral = $\frac{1}{2}(ab + cd) \sin B$.
- (4) Area of cyclic quadrilateral = $\sqrt{(s-a)(s-b)(s-c)(s-d)}$.
- (5) $\cos B = \frac{a^2 + b^2 - c^2 - d^2}{2(ab + cd)}$
- (6) $\cos D = \frac{c^2 + d^2 - a^2 - b^2}{2(ab + cd)}$.
- (7) Circum-radius = $\frac{1}{4\Delta} \sqrt{(ac + bd)(ad + bc)(ab + cd)} = \frac{1}{4} \sqrt{\frac{(ac + bd)(ad + bc)(ab + cd)}{(s-a)(s-b)(s-c)(s-d)}}$.

(bb) **Regular Polygon:** A regular polygon is a polygon which has all its sides equal and all angles are equal.



- (1) Sum of each interior angles of a regular polygon = $\pi(n-2)$.
- (2) Each interior angle of a regular polygon of n sides = $\frac{\pi(n-2)}{n}$.
- (3) Each exterior angles of a regular polygon of n sides = $\frac{2\pi}{n}$.
- (4) In-radius of the regular polygon = $r = \frac{1}{2}a \cot \frac{\pi}{n}$.
- (5) Circum-radius of the regular polygon = $R = \frac{1}{2}a \operatorname{cosec} \frac{\pi}{n}$.
- (6) Area of regular polygon = $n \times \text{Area of triangle OAB}$
 $\Rightarrow \text{Area of regular polygon} = \frac{1}{4}na^2 \cot \frac{\pi}{n}$ (in terms of side 'a').
 $\Rightarrow \text{Area of regular polygon} = nr^2 \tan \frac{\pi}{n}$ (in terms of in-radius 'r').
 $\Rightarrow \text{Area of regular polygon} = \frac{1}{2}nR^2 \sin \frac{2\pi}{n}$ (in terms of circum-radius 'R').

Mensuration

1. Area Surface & Volume

Triangle:

- Perimeter of a triangle = Sum of all the three sides.
- Area of a triangle = $\frac{1}{2} \times \text{Base} \times \text{Height}$.
- Area of a triangle = $\sqrt{s(s-a)(s-b)(s-c)}$, where $s = \frac{a+b+c}{2}$.
- Perimeter of equilateral triangle = $3 \times \text{side}$.
- Area of equilateral triangle = $\frac{\sqrt{3}}{4} \times (\text{side})^2 = \frac{1}{\sqrt{3}} \times (\text{height})^2$.
- Height of equilateral triangle = $\frac{\sqrt{3}}{2} \times \text{side}$.
- Side of equilateral triangle = $\frac{2}{\sqrt{3}} \times \text{Height}$.
- Area of isosceles triangle = $\frac{1}{2} \times \text{Base} \times \sqrt{(\text{equal side})^2 - \frac{(\text{Base})^2}{4}}$.
- Height of isosceles triangle = $\sqrt{(\text{equal side})^2 - \frac{(\text{Base})^2}{4}}$.
- Base of isosceles triangle = $2\sqrt{(\text{equal side})^2 - (\text{height})^2}$.
- Equal side isosceles triangle = $\sqrt{(\text{height})^2 + \frac{(\text{Base})^2}{4}}$.
- Hypotenuse of right angled isosceles triangle = $\sqrt{2} \times \text{side}$.
- Equal Side of right angled isosceles triangle = $\frac{\text{Hypotenuse}}{\sqrt{2}}$.
- Area of right angled isosceles Triangle = $\frac{1}{2} \times (\text{equal side})^2 = \frac{1}{4} \times (\text{Hypotenuse})^2$.

Square:

- Perimeter of a square = $4 \times \text{side}$
- Area of square = $(\text{side})^2$.
- Diagonal of a square = $\sqrt{2} \times \text{side}$.

Rectangle:

- Perimeter of rectangle = $2 \times (\text{length} + \text{breadth})$.
- Area of rectangle = $\text{length} \times \text{breadth}$.
- Diagonal of rectangle = $\sqrt{(\text{length})^2 + (\text{breadth})^2}$.

- Area of Four Walls = $2 \times \text{height} \times (\text{length} + \text{breadth})$.

Parallelogram:

- Area of a parallelogram = $\text{Base} \times \text{Height}$.

Rhombus:

- Area of rhombus = $\frac{1}{2} \times \text{Product of diagonals}$.
- Side of a rhombus = $\frac{1}{2} \sqrt{(d_1)^2 + (d_2)^2}$, where d_1, d_2 are lengths of diagonals.

Trapezium:

- Area of trapezium = $\frac{1}{2} \times \text{Sum of parallel sides} \times \text{Height}$

Circle:

- Circumference of a circle = $2\pi.r = \pi.d$.
- Area of circle = $\pi.r^2 = \frac{\pi.d^2}{4}$.
- Radius of circle = $\frac{\text{diameter}}{2}$.

Arc:

- Length of arc = $\frac{\theta}{360^\circ} \times 2\pi.r$.
- Area of arc = $\frac{\theta}{360^\circ} \times \pi.r^2$

Cube: Let side of the cube = a .

- Surface area of cube = $6a^2$.
- Area of four walls of cube = $4a^2$.
- Volume of the cube = a^3 .
- Main diagonal of a cube = $\sqrt{3} \times a$.

Cuboid: Let length = l , breadth = b and height = h

- Total surface area of cuboid = $2(lb + bh + hl)$
- Area of four walls of cuboid = $2h(l + b)$.
- Volume of the cuboid = $l \times b \times h$.
- Main diagonal of a cuboid = $\sqrt{l^2 + b^2 + h^2}$.

Right Circular Cylinder: Let radius of base = r and height = h

- Base area of cylinder = $\pi.r^2$
- Curved surface area of cylinder = $2\pi.rh$
- Total surface area of cylinder = $2\pi.rh + 2\pi.r^2 = 2\pi.r(r + h)$.
- Volume of cylinder = $\pi.r^2h$

Right Circular Cone: Let radius of base = r , height = h and slant height = l

- Slant height = $l = \sqrt{r^2 + h^2}$
- Base area of cone = $\pi.r^2$
- Curved surface area of cone = $\pi.rl$
- Total Surface area of cone = $\pi.r^2 + \pi.rl = \pi.r(r + l)$.
- Volume of the cone = $\frac{1}{3}\pi.r^2h$

Sphere:

- Curved surface area of sphere = $4\pi.r^2$
- Volume of the sphere = $\frac{4}{3}\pi.r^3$.

Hemisphere:

- Curved surface area of hemisphere = $2\pi.r^2$
- Base area of hemisphere = $\pi.r^2$
- Total surface area of hemisphere = $3\pi.r^2$
- Volume of the hemisphere = $\frac{2}{3}\pi.r^3$.

Frustum: Let h be height, l be the slant height and R and r are the radii of both the ends of a frustum.

- Slant height of frustum = $l = \sqrt{h^2 + (R - r)^2}$
- Smaller base area of frustum = $\pi.r^2$
- Bigger base area of frustum = $\pi.R^2$
- Curved surface area of frustum = $\pi.(R + r)l$.
- Total surface area of frustum = $\pi.r^2 + \pi.R^2 + \pi.(R + r)l = \pi.(r^2 + R^2 + Rl + rl)$.
- Volume of frustum = $\frac{1}{3}\pi.h(R^2 + r^2 + Rr)$.

n-Gon: Let a n-gon is there.

- Sum of interior angles of n-gon = $(n - 2) \times 180^\circ$.
- Sum of Exterior angles of n-gon = 360°
- For a regular n-gon each interior angles = $\frac{(n - 2)}{n} \times 180^\circ$.

- For a regular n-gon each exterior angles = $\frac{360^0}{n}$.

Exercise:

1. P is any point inside a triangle ABC. The perimeter of the triangle $AB + BC + CA = 2s$. Prove that $s < AP + BP + CP < 2s$.
2. If the circum centre and centroid of a triangle coincide, prove that the triangle must be equilateral.
3. Take any point P_1 on side of BC of a triangle and draw following chain of lines; P_1P_2 parallel to AC, P_2P_3 parallel to BC; P_3P_4 parallel to AB; P_4P_5 parallel to CA; P_5P_6 parallel to BC. Here P_2, P_5 lie on AB; P_3, P_6 lie on CA and P_4 lie on BC. Show that P_6P_1 is parallel to AB.
4. Let P be an interior point of a triangle ABC and AP, BP, CP meets the sides BC, CA and AB in point D, E and F respectively. Show that $\frac{AP}{PD} = \frac{AF}{FB} + \frac{AE}{EC}$.
5. ABCD is a cyclic quadrilateral with $AC \perp BD$; AC meets BD at E. Prove that $EA^2 + EB^2 + EC^2 + ED^2 = 4R^2$, where R is the radius of circumscribing circle.
6. ABCD is a cyclic quadrilateral; x, y, z are the distances of A from the lines BD, BC and CD respectively. Prove that $\frac{BD}{x} = \frac{BC}{y} + \frac{CD}{z}$.
7. ABCD is a quadrilateral and P, Q are midpoints of CD and AB. AP, DQ meets at X, and BP, CQ meets at Y. Prove that area $\triangle ADX + \text{area } \triangle BCY = \text{area quadrilateral } PXQY$.
8. The cyclic octagon ABCDEFGH has sides a, a, a, a, b, b, b, b respectively. Find the radius of the circle that circumscribe ABCDEFGH.
9. Let ABC be an acute angled triangle and CD be the altitude through c. If $AB = 8$, $CD = 6$, find the distance between the mid points of AD and BC.
10. In a triangle ABC, the incircle touches the sides BC, CA and AB respectively at D, E and F. If the radius of the circle is 4 units and if BD, CE and AF are consecutive integers, find the sides of the triangle ABC.
11. Let AC and BD be two chords of a circle with centre O such that they intersect at right angles inside the circle at the point M. Suppose K and L are mid points of the chord AB and CD respectively. Prove that OKML is a parallelogram.
12. In triangle ABC, K and L are points on the sides BC (K being closer to B than L) such that $BC \cdot KL = BK \cdot CL$ and AL bisect $\angle KAC$. Show that AL is perpendicular to AB.
13. The sides of a triangle are three consecutive integers and its inradius is four units. Determine the circum radius.
14. Let P be an interior point of a triangle ABC and let BP and CP meet AC and AB in E and F respectively. If $[BPF] = 4$, $[BPC] = 8$, and $[CPE] = 13$, find $[AFPE]$, where $[]$ denotes area.
15. Let ABC be a right angled triangle with $\angle B = 90^0$ and let BD be the altitude from B on AC. Draw $DE \perp AB$ and $DF \perp BC$. Let P, Q, R and S be respectively the incentres of triangle DFC, DBF, DEB and DAE. Suppose S, R, Q are collinear. Prove that P, Q, R, D lie on a circle.

16. Two circles Γ and Σ in the plane intersect at two distinct points A and B. A line through B intersects Γ and Σ again in C and D, respectively. Suppose $CA = CD$. Show that the centre of Σ lies on Γ .
17. Let ABCD be an isosceles trapezium having an incircle; let AB and CD be the parallel sides and let CE be the perpendicular from C on to AB. Prove that CE is equal to the geometric mean of AB and CD.
18. What is the minimal area of a right angled triangle whose in-radius is 1 unit?
19. Let Γ be a circle with centre O. Let Λ be another circle passing through O and intersecting Γ at points A and B. A diameter CD of Γ intersects Λ at a point P different from O. Prove that $\angle APC = \angle BPD$.
20. Let ABC be a triangle. Let E be a point on the segment BC such that $BE = 2EC$. Let F be the mid point of AC. Let BF intersect AE in Q. Determine BQ/QF.
21. Let ABCD be a unit square. Draw a quadrant of a circle with A as centre and B, D as end points of the arc. Similarly draw a quadrant of a circle with B as centre and A, C as end points of the arc. Inscribe a circle Γ touching the arc AC internally, the arc BD internally and also touching the side AB. Find the radius of the circle Γ .
22. Let ABC be a triangle. Let D, E, F be points respectively on the segments BC, CA, AB such that AD, BE, CF concur at point K. Suppose $\frac{BD}{DC} = \frac{BF}{FA}$ and $\angle ADB = \angle AFC$. Prove that $\angle ABE = \angle CAD$.
23. Let ABC be a triangle and let BB_1, CC_1 be respectively the bisector of $\angle B, \angle C$ with B_1 on AC and C_1 on AB. Let E, F be the feet of perpendiculars drawn from A onto BB_1 and CC_1 respectively. Suppose D is the point at which the incircle of ABC touches AB. Prove that $AD = EF$.
24. Let ABC be a triangle in which $m\angle A = 60^\circ$. Let BE and CF are bisectors of $\angle B$ and $\angle C$ with E on AC and F on AB. Let M be the reflection of A in the line EF. Prove that M lies on BC.

QUADRATIC EXPRESSION

- **Polynomial**: - Any expression $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$, where $a_0, a_1, a_2, \dots, a_n$ are constants and $a_n \neq 0$, is called as a polynomial of degree n .
- **Equation**: - An expression of equality is known as equation or an expression equals to another expression is known as equation.
- **Algebraic Equation**: - If any equation involves polynomial in one variable, then it is called as algebraic equation.
- **Transcendental Equation**: - If any equation involves trigonometric, logarithmic, exponential expression, then it is called as transcendental equation.
- **Different Types of Algebraic Equation**: -
 5. **Linear Equation**: - An equation which contains linear polynomial is called as linear equation.
Example: - $ax + b = 0$
 6. **Quadratic Equation**: - An equation which contains quadratic polynomial is called as quadratic equation.
Example: - $ax^2 + bx + c = 0$
 7. **Cubic Equation**: - An equation which contains cubic polynomial is called as cubic equation.
Example: - $ax^3 + bx^2 + cx + d = 0$
 8. **Biquadratic Equation**: - An equation which contains biquadratic polynomial is called as biquadratic equation.
Example: - $ax^4 + bx^3 + cx^2 + dx + e = 0$
- **Solution of Algebraic Equation**: - For any value of x which satisfies $f(x) = 0$ are called as solution of $f(x) = 0$ or root of $f(x) = 0$.
Example: - For $2x + 6 = 0$
 - c. If $x = 2$, it is not satisfies $2x + 6 = 0$. So, $x = 2$ is not a solution of $2x + 6 = 0$.
 - d. If $x = -3$, it is not satisfies $2x + 6 = 0$. So, $x = -3$ is a solution of $2x + 6 = 0$.
- **General Properties of roots of Equation**: -
 8. If α is a root of the equation $f(x) = 0$, then $f(x)$ is exactly divisible by $x - \alpha$ or $x - \alpha$ is a factor of $f(x)$.

9. If $f(x)$ is exactly divisible by $x - \alpha$, then α is a root of $f(x) = 0$.
10. Every equation of n th degree has exactly n roots, may be real, and may be imaginary.
11. Let $f(x) = 0$ such that, $f(a)$ and $f(b)$ have different sign then equation $f(x) = 0$ has at least one root lies between a and b .
12. In an equation with real coefficients, if $\alpha + i\beta$ is a root then $\alpha - i\beta$ is also one of the root. Or imaginary roots always occur in conjugate pairs.
13. If $f(x) = 0$ has $a + \sqrt{b}$ is one of the root, then $a - \sqrt{b}$ is also another root.
14. Every equation of odd degree has atleast one real root.

• **Descarte's Rule of Sign:-**

- d. Maximum number of positive real root of $f(x) = 0$ is same as the number of sign changes in $f(x)$.
 - e. Maximum number of negative real root of $f(x) = 0$ is same as the number of sign changes in $f(-x)$.
- If an equation having degree n such that at most p number of positive roots, at most q number of negative roots, then equation has atleast $n - (p + q)$ number of imaginary roots.
 - **Quadratic Equation:** An equation of the form $ax^2 + bx + c = 0$ where $a \neq 0, b, c$ are coefficients is called as quadratic equation.
 - The roots of the quadratic equation $ax^2 + bx + c = 0$ are given by $x = \frac{-b \pm \sqrt{D}}{2a}$ where $D = b^2 - 4ac$ is called as discriminant of quadratic equation.
 - The quadratic equation whose two roots α, β are given is $\Rightarrow x^2 - (\text{sum of roots})x + \text{product of roots} = 0 \Rightarrow x^2 - (\alpha + \beta)x + \alpha\beta = 0$.
 - If α, β are the roots of the equation $ax^2 + bx + c = 0$ then:
 1. Sum of roots $= \alpha + \beta = -\frac{b}{a} = -\frac{\text{coefficient of } x}{\text{coefficient of } x^2}$.
 2. Product of roots $= \alpha\beta = \frac{c}{a} = \frac{\text{constant term}}{\text{coefficient of } x^2}$.
 3. $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = \left(-\frac{b}{a}\right)^2 - 2\frac{c}{a} = \frac{b^2}{a^2} - \frac{2c}{a} = \frac{b^2 - 2ac}{a^2}$.
 4. $|\alpha - \beta| = \sqrt{(\alpha + \beta)^2 - 4\alpha\beta} = \sqrt{\left(-\frac{b}{a}\right)^2 - 4\frac{c}{a}} = \sqrt{\frac{b^2}{a^2} - \frac{4c}{a}} = \sqrt{\frac{b^2 - 4ac}{a^2}} = \frac{\sqrt{b^2 - 4ac}}{|a|}$.
 - In the quadratic equation $ax^2 + bx + c = 0$,
 1. If a and c are of same sign then, product of roots $\frac{c}{a}$ is positive.
 2. If a and c are of opposite sign then, product of roots $\frac{c}{a}$ is negative.

3. If $a + b + c = 0$ then roots are 1 and $\frac{c}{a}$.
 4. If $a - b + c = 0$ then roots are -1 and $-\frac{c}{a}$.
 5. If the roots are negative, then a, b, c will have same sign.
 6. If the roots are positive, then a, c will have same sign different from sign of b .
- The roots of $ax^2 + bx + a = 0$ are reciprocal to each other.
 - **Nature of roots:** In the quadratic equation $ax^2 + bx + c = 0$, coefficients a, b, c are rationals:
 1. If $D = b^2 - 4ac > 0$ then roots are real and distinct.
 2. If $D = b^2 - 4ac > 0$ and is a perfect square then roots are rational and unequal.
 3. If $D = b^2 - 4ac > 0$ and is a not perfect square then roots are irrational and unequal.
 4. If $D = b^2 - 4ac = 0$ then roots are rational and equal.
 5. If $D = b^2 - 4ac < 0$ then roots are imaginary and conjugate to each other.
 6. If $ax^2 + bx + c = 0$ has more than two roots, then this is an identity i.e. $a = b = c$.
 - **Nature of the Graph of $f(x) = ax^2 + bx + c$, $a \neq 0, b, c \in R$:**
 - (1) The shape of the curve is a parabola.
 - (2) Y-axis is the axis of parabola.
 - (3) If $a > 0$, the parabola opens upwards.
 - (4) If $a < 0$, the parabola opens downwards.
 - (5) If $D = b^2 - 4ac > 0$, the parabola cuts x-axis at two points.
 - (6) If $D = b^2 - 4ac = 0$, the parabola touch x-axis at one points and the co-ordinate of vertex of parabola is $\left(\frac{-b}{2a}, \frac{-D}{4a}\right)$.
 - (7) Length of latus rectum is $\frac{1}{|a|}$.
 - (8) If $D = b^2 - 4ac < 0$, the parabola does not cut x-axis at any points.
 - **Greatest & Least Value of Quadratic Expression:**
 - (1) If $a > 0$, then the expression $f(x) = ax^2 + bx + c$, $a \neq 0, b, c \in R$ has no greatest value but it has least value $\frac{4ac - b^2}{4a}$ at $x = \frac{-b}{2a}$.
 - (2) If $a < 0$, then the expression $f(x) = ax^2 + bx + c$, $a \neq 0, b, c \in R$ has no least value but it has greatest value $\frac{4ac - b^2}{4a}$ at $x = \frac{-b}{2a}$.

- **Sign of Quadratic Expression:** Let α, β are the roots of quadratic expression $f(x) = ax^2 + bx + c = 0$. So at $x = \alpha$, $x = \beta$ the value of expression $ax^2 + bx + c$ is zero and for other values of x the value of expression $ax^2 + bx + c$ is positive or negative as per the following:

- (1) If α, β are the real and distinct roots of $f(x) = ax^2 + bx + c = 0$. i.e. ($D > 0$), then $ax^2 + bx + c > 0$ if $x < \alpha$ or $x > \beta$.
- (2) If α, β are the real and distinct roots of $f(x) = ax^2 + bx + c = 0$. i.e. ($D > 0$), then $ax^2 + bx + c < 0$ if $\alpha < x < \beta$.
- (3) If α, β are the real and equal roots of $f(x) = ax^2 + bx + c = 0$. i.e. ($D = 0$), then $ax^2 + bx + c \geq 0$ if $a > 0$.
- (4) If α, β are the real and equal roots of $f(x) = ax^2 + bx + c = 0$. i.e. ($D = 0$), then $ax^2 + bx + c \leq 0$ if $a < 0$.
- (5) If α, β are the imaginary roots of $f(x) = ax^2 + bx + c = 0$. i.e. ($D < 0$), then $ax^2 + bx + c > 0$ if $a > 0$.
- (6) If α, β are the imaginary roots of $f(x) = ax^2 + bx + c = 0$. i.e. ($D < 0$), then $ax^2 + bx + c < 0$ if $a < 0$.

- **Position of roots of a quadratic equation:** Let $f(x) = ax^2 + bx + c$, $a \neq 0, b, c \in R$ be a quadratic expression.

- (1) **Conditions for Both roots to be more than a Real number k :**

- (a) $D > 0$.
- (b) $k < \frac{-b}{2a}$.
- (c) $af(k) > 0$.

- (2) **Conditions for Both roots to be less than a Real number k :**

- (a) $D > 0$.
- (b) $k > \frac{-b}{2a}$.
- (c) $af(k) > 0$.

- (3) **Conditions for a Real number k lies between Both roots:**

- (a) $af(k) < 0$.

- (4) **Conditions for exactly one root to lie in between the interval (k_1, k_2) :**

- (a) If $f(k_1)f(k_2) < 0$

(5) **Conditions for both the root to lie in between the interval** (k_1, k_2) :

(a) If $a > 0$, $D > 0$, $k_1 < \frac{-b}{2a} < k_2$ and $f(k_1) > 0$ & $f(k_2) > 0$.

(b) If $a < 0$, $D > 0$, $k_1 < \frac{-b}{2a} < k_2$ and $f(k_1) < 0$ & $f(k_2) < 0$

- **Conditions for a quadratic equation to have a repeated roots:** The quadratic equation $f(x) = ax^2 + bx + c = 0$, $a \neq 0, b, c \in R$ has α as a repeated roots if and only if $f(\alpha) = 0$ and $f'(\alpha) = 0$ and the repeated roots are $\alpha = \frac{-b}{2a}$.

- **Condition for Two quadratic equations to have a common roots:** Two quadratic equation $a_1x^2 + b_1x + c_1 = 0$ and $a_2x^2 + b_2x + c_2 = 0$ have a common root if $(a_1b_2 - a_2b_1)(b_1c_2 - b_2c_1) = (c_1a_2 - c_2a_1)^2$.

- **Condition for Two quadratic equations to have equal roots:** Two quadratic equation $a_1x^2 + b_1x + c_1 = 0$ and $a_2x^2 + b_2x + c_2 = 0$ have equal root if $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$.

- **Note point:**

- (1) To find the common roots of the two quadratic equations, make the coefficient of second degree terms in two equations equal and subtract. The values of x so obtained are the required common root.
- (2) If two quadratic equations with real coefficients have an imaginary roots common, then both roots will be common and the two equations will be identical with $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$.
- (3) If two quadratic equations with rational coefficients have an irrational roots common, then both roots will be common and the two equations will be identical with $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$.
- (4) If α is a repeated roots of quadratic equation $f(x) = ax^2 + bx + c = 0$, $a \neq 0, b, c \in R$, then α is a root of $f'(x) = 0$.

- **Higher Degree Equation:**

- **Relation between Roots and Coefficients of an equation:** -

d. For the quadratic equation $a_0x^2 + a_1x + a_2 = 0$. If α_1, α_2 are two roots then

$$\sum \alpha_i = \alpha_1 + \alpha_2 = -\frac{a_1}{a_0}$$

$$\alpha_1 \cdot \alpha_2 = \frac{a_2}{a_0}$$

- e. For the cubic equation $a_0x^3 + a_1x^2 + a_2x + a_3 = 0$. If $\alpha_1, \alpha_2, \alpha_3$ are three roots then

$$\sum \alpha_i = \alpha_1 + \alpha_2 + \alpha_3 = -\frac{a_1}{a_0}$$

$$\sum \alpha_i \alpha_j = \alpha_1 \alpha_2 + \alpha_2 \alpha_3 + \alpha_3 \alpha_1 = \frac{a_2}{a_0}$$

$$\alpha_1 \alpha_2 \alpha_3 = -\frac{a_3}{a_0}$$

- f. Similarly, Let us consider nth degree equation $a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_n = 0$. Let $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n$ are n numbers of roots then

$$\sum \alpha_i = -\frac{a_1}{a_0}$$

$$\sum \alpha_i \alpha_j = \frac{a_2}{a_0}$$

$$\sum \alpha_i \alpha_j \alpha_k = -\frac{a_3}{a_0} \dots\dots\dots$$

$$\alpha_1 \alpha_2 \alpha_3 \dots \alpha_n = (-1)^n \frac{a_n}{a_0}$$

• **Transformation of Equations:**

- (1) If $f(x) = 0$ is a quadratic equation, then equation whose roots are reciprocal of roots of $f(x) = 0$ is $f\left(\frac{1}{x}\right) = 0$.
- (2) If $f(x) = 0$ is a quadratic equation, then equation whose roots are increased by k than the roots of $f(x) = 0$ is $f(x - k) = 0$.
- (3) If $f(x) = 0$ is a quadratic equation, then equation whose roots are decreased by k than the roots of $f(x) = 0$ is $f(x + k) = 0$.
- (4) If $f(x) = 0$ is a quadratic equation, then equation whose roots are negative of the roots of $f(x) = 0$ is $f(-x) = 0$.
- (5) If $f(x) = 0$ is a quadratic equation, then equation whose roots are k times the roots of $f(x) = 0$ is $f\left(\frac{x}{k}\right) = 0$.
- (6) If $f(x) = 0$ is a quadratic equation, then equation whose roots are squares of the roots of $f(x) = 0$ is $f(\sqrt{x}) = 0$.
- (7) If $f(x) = 0$ is a quadratic equation, then equation whose roots are cubes of the roots of $f(x) = 0$ is $f(\sqrt[3]{x}) = 0$.
- (8) If α, β are the roots of the equation $f(x) = 0$, then equation whose roots are $a\alpha + b, a\beta + b$ is $f\left(\frac{x - b}{a}\right) = 0$.

• **Note Points:**

- (1) Condition for the roots of $ax^2 + bx + c = 0$ are in the ratio $m : n$ is $(m+n)^2 ac = mnb^2$.
- (2) Condition for one of the root of $ax^2 + bx + c = 0$ is square of other is $b^3 + a^2c + ac^2 = 3abc$.
- (3) Condition for one of the root of $ax^2 + bx + c = 0$ is n th power of other is $(ac^n)^{\frac{1}{n+1}} + (a^n c)^{\frac{1}{n+1}} + b = 0$.
- (4) If the difference of the roots of $ax^2 + bx + c = 0$ is same as the difference of the roots of $px^2 + qx + r = 0$, $\frac{b^2 - 4ac}{q^2 - 4pr} = \frac{a^2}{p^2}$.
- (5) In a quadratic equation with rational coefficients irrational roots occurs as conjugate pairs.
- (6) In a quadratic equation with rational coefficients imaginary roots occurs as conjugate pairs.
- (7) The equation $a_1x^2 + b_1x + c_1 = 0$ and $a_2x^2 + b_2x + c_2 = 0$ where $a_1b_2 - a_2b_1 \neq 0$ have a common root if $(c_1a_2 - c_2a_1)^2 = (a_1b_2 - a_2b_1)(b_1c_2 - b_2c_1)$ and the common roots is $\frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1}$ or $\frac{b_1c_2 - b_2c_1}{c_1a_2 - c_2a_1}$.
- (8) If the equation $a_1x^2 + b_1x + c_1 = 0$ and $a_2x^2 + b_2x + c_2 = 0$ have common roots then $a_1 : a_2 = b_1 : b_2 = c_1 : c_2$.
- (9) If $f(x) = \frac{ax^2 + bx + c}{ax^2 - bx + c}$ or $f(x) = \frac{ax^2 - bx + c}{ax^2 + bx + c}$ and $b^2 - 4ac < 0$, then minimum and maximum value of $f(x)$ are given by $f\left(\pm \sqrt{\frac{c}{a}}\right)$.
- (10) If $f(x) = \frac{(x+a)(x+b)}{x+c}$, then minimum value of $f(x)$ is $(\sqrt{a-c} + \sqrt{b-c})^2$ and it occurs at $-c + \sqrt{(a-c)(b-c)}$ where $a > c, b > c$.
- (11) If $f(x) = \frac{(x+a)(x+b)}{x+c}$, then maximum value of $f(x)$ is $(\sqrt{a-c} - \sqrt{b-c})^2$ and it occurs at $-c - \sqrt{(a-c)(b-c)}$ where $a < c, b < c$.
- (12) An expression of the form $ax^2 + 2hxy + by^2$ where $a, b, h \in C$ is called as homogeneous expression of second degree in x, y .
- (13) An expression of the form $ax^2 + 2hxy + by^2 + 2gx + 2fy + c$ where $a, b, c, f, g, h \in C$ and atleast one of where a, b, h is not equal to zero is called a general expression of second degree in x, y .
- (14) The necessary and sufficient condition for second degree expression $ax^2 + 2hxy + by^2 + 2gx + 2fy + c$ to be expressed as a product of two linear factors is $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$ or $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0, h^2 \geq ab, g^2 \geq ac, f^2 \geq bc$.

- (15) If $f(x) = 0$ be any equation and a and b are two real numbers such that $f(a)f(b) < 0$, then equation $f(x) = 0$ has atleast one real root or an odd number of real roots (may be repeated) lies between a and b .
- (16) If $f(x) = 0$ be any equation and a and b are two real numbers such that $f(a)f(b) > 0$, then equation $f(x) = 0$ has either no real roots or an even number of real roots (may be repeated) lies between a and b .

• **Solution of In-equation:**

Here $a > 0$

- (1) $|x| < a \Leftrightarrow -a < x < a$.
- (2) $|x| > a \Leftrightarrow -a < x \cup x > a$.
- (3) If $\alpha < \beta$, $(x - \alpha)(x - \beta) < 0 \Leftrightarrow \alpha < x < \beta$.
- (4) If $\alpha < \beta$, $(x - \alpha)(x - \beta) \leq 0 \Leftrightarrow \alpha \leq x \leq \beta$.
- (5) If $\alpha < \beta$, $(x - \alpha)(x - \beta) > 0 \Leftrightarrow x < \alpha \cup x > \beta$.
- (6) If $\alpha < \beta$, $(x - \alpha)(x - \beta) \geq 0 \Leftrightarrow x \leq \alpha \cup x \geq \beta$.

• **(Wavy Curve Method) To find Method of Intervals where the rational function is positive or negative:**

Let $f(x) = \frac{g(x)}{h(x)} = \frac{(x - a_1)^{m_1}(x - a_2)^{m_2} \dots (x - a_k)^{m_k}}{(x - b_1)^{n_1}(x - b_2)^{n_2} \dots (x - b_r)^{n_r}}$ is > 0 , < 0 , ≥ 0 , ≤ 0 such that $a_i \neq b_j$ $\left(\begin{matrix} i = 1, 2, \dots, k \\ j = 1, 2, \dots, r \end{matrix} \right)$, m_i, n_j $\left(\begin{matrix} i = 1, 2, \dots, k \\ j = 1, 2, \dots, r \end{matrix} \right)$ are natural numbers.

Steps to find sign of the function in the intervals:

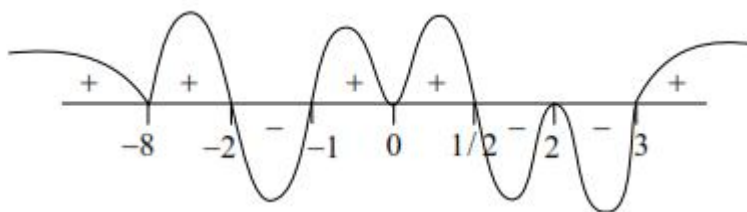
- (1) Find all zeroes of $g(x)$ and $h(x)$ and marked in the number line, they are called as critical points.
- (2) All zeroes of $h(x)$ are known as points of discontinuities.
- (3) $g(x)$ and $h(x)$ should not contain common zeroes.
- (4) Even and odd powers of all the zeroes of $g(x)$ and $h(x)$ should be noted.
- (5) Now after marking the numbers on the number line put (+ve) sign in the right of the biggest of these numbers.
- (6) When passing through the next number the polynomial changes its sign if its power is odd and have the same sign if its power is even and continue the process.
- (7) The solution of $f(x) > 0$ is the union of all those intervals in which we have put (+ve) sign.
- (8) The solution of $f(x) < 0$ is the union of all those intervals in which we have put (-ve) sign.

Example: $f(x) = \frac{(x - 2)^{442}(x + 1)^{123}(x - 0.5)^{91}(x + 8)^{50}}{x^{92}(x - 3)^{87}(x + 2)^{41}}$

Solution: Here critical points are -8, -2, -1, 0, 0.5, 2, 3

-8	Even Powers
-2	Odd Powers

- 1 Odd Powers
- 0 Even Powers
- 0.5 Odd Powers
- 2 Even Powers
- 3 Odd Powers



Here $f(x) > 0$ for $(-\infty, -2) \cup (-1, 0.5) \cup (3, \infty)$.

And $f(x) < 0$ for $(-2, -1) \cup (0.5, 3)$.

• **Equation Reducible to Linear, Quadratic and Biquadratic Equations:**

- (1) To solve equation of the type: $(x-a)^4 + (x-b)^4 = A$ put $y = x - \frac{a+b}{2}$.
- (2) To solve equation of the type: $(x-a)^{2n} + (x-b)^{2n} = A$ put $y = x - \frac{a+b}{2}$.
- (3) To solve equation of the type: $a_0(f(x))^{2n} + a_1(f(x))^n + a_2 = 0$ put $(f(x))^n = y$.
- (4) To solve equation of the type: $(ax^2 + bx + c_1)(ax^2 + bx + c_2) \dots (ax^2 + bx + c_n) = A$ put $ax^2 + bx = y$.
- (5) To solve equation of the type: $(x-a)(x-b)(x-c)(x-d) = Ax^2$, where $ab = cd$ put $y = x + \frac{ab}{x}$.
- (6) To solve equation of the type: $(x-a)(x-b)(x-c)(x-d) = A$, where $b-a = d-c$, put $y = x - \frac{a+b+c+d}{4}$ or $y = \frac{(x-a)+(x-b)+(x-c)+(x-d)}{4}$.
- (7) A polynomial $f(x, y)$ is said to be symmetric if $f(x, y) = f(y, x) \forall x, y$.
- (8) Any symmetric polynomial can be represented in the form of $x+y$ or xy .

• **Use of Continuity & Differentiability to find roots of a polynomial:**

Let $f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_n$, then $f(x)$ is continuous on $\mathbb{R}$.

As $f(x)$ is continuous on $\mathbb{R}$, we can use intermediate value theorem to find whether $f(x)$ has real roots or not. If there exist a and b such that $f(a)f(b) < 0$, then there exist atleast one $c \in (a, b)$ such that $f(c) = 0$. Also if $f(-\infty)$ and $f(a)$ are of opposite sign, then atleast one root lies between $(-\infty, a)$. If $f(a)$ and $f(\infty)$ are of opposite sign, then atleast one root lies between (a, ∞) .

- (1) If $f(x) = 0$ has a root α of multiplicity $r (r > 1)$, then we have $f(x) = (x - \alpha)^r g(x)$, where $g(\alpha) \neq 0$. Also $f'(x) = 0$ has α as a root with multiplicity $r - 1$.
- (2) If $f(x) = 0$ has n real roots, then $f'(x) = 0$ has at least $n - 1$ real roots.
- (3) If $f(x) = 0$ has n distinct real roots, then we can write :

$$f(x) = a_0(x - a_1)(x - a_2) \dots (x - a_n).$$

$$\text{Then } \frac{f'(x)}{f(x)} = \sum_{k=1}^n \frac{1}{x - a_k}.$$

• **Cardan's Method Solution of Cubic Equation:**

Let us solve $ax^3 + bx^2 + cx + d = 0$ by Cardan's Method

1. Divide equation (1) by coefficient of x^3 i.e. by a , then equation (1) becomes
 $x^3 + lx^2 + mx + n = 0$ ----- $\rightarrow (2)$
2. Remove x^2 term from equation (2) by putting $y = x + \frac{l}{3} \Rightarrow x = y - \frac{l}{3}$. Then equation (2) becomes $y^3 + py + q = 0$ ----- $\rightarrow (3)$
3. Put $y = u + v$ in equation (3), we get
 $\Rightarrow y^3 = u^3 + v^3 + 3uv(u + v) = u^3 + v^3 + 3uvy$
 $\Rightarrow y^3 - 3uvy - (u^3 + v^3) = 0$ ----- $\rightarrow (4)$
4. From comparing equation (3) and (4) we get
 $uv = \frac{-p}{3}$ and $u^3 + v^3 = -q$
 $\Rightarrow u^3 + v^3 = -q, \quad u^3 v^3 = \frac{-p^3}{27}$
5. u^3, v^3 are the roots of $t^2 - (u^3 + v^3)t + u^3 v^3 = 0$.
 $\Rightarrow t^2 + qt - \frac{p^3}{27} = 0$
 $\Rightarrow t = \frac{1}{2} \left(-q \pm \sqrt{q^2 + \frac{4p^3}{27}} \right)$
 So, $\Rightarrow u^3 = \frac{1}{2} \left(-q + \sqrt{q^2 + \frac{4p^3}{27}} \right), \quad v^3 = \frac{1}{2} \left(-q - \sqrt{q^2 + \frac{4p^3}{27}} \right)$
 Then the values of u are $\lambda, \quad \lambda\omega, \quad \lambda\omega^2$
6. As $uv = \frac{-p}{3}$.
 So, corresponding values of v are $\frac{-p}{3\lambda}, \quad \frac{-p}{3\lambda\omega}, \quad \frac{-p}{3\lambda\omega^2}.$

7. Then find the values of y by putting the values of u and v in $y = u + v$.
8. Then finally find the values of x by putting the values of y in $x = y - \frac{l}{3}$.

• **Ferrari's Method Solution of Biquadratic Equation:**

Let us solve $x^4 + ax^3 + bx^2 + cx + d = 0$ by Ferrari's method.-----→(1)

1. By combining x^4 and x^3 terms into a perfect square equation (1) can be written as

$$\left(x^2 + \frac{a}{2}x + \lambda\right)^2 - (mx + n)^2 = 0 \text{ .-----}(2)$$

$$\begin{array}{lcl} \text{Where } 2\lambda - m^2 + \frac{a^2}{4} = b & \Rightarrow m^2 = 2\lambda + \frac{a^2}{4} - b & \\ a\lambda - 2mn = c & \Rightarrow 2mn = a\lambda - c & \\ \lambda^2 - n^2 = d & \Rightarrow n^2 = \lambda^2 - d & \end{array} \quad \text{-----} \rightarrow (3)$$

2. Eliminate m and n from (3)
i.e. $m^2 \times n^2 = (mn)^2$ and find the value of λ .
3. Then put value of λ in equation (3) to find m and n .
4. Next put value of λ , m and n in (2) and find the values of x .

MATHEMATICAL INDUCTION

• **Mathematical Induction Principle:** Let $P(n)$ be any statement where $n \in N$,

- (a) Prove that $P(1)$ is true.
- (b) Let us assume that $P(k)$ be true, $k \leq n \in N$.
- (c) Prove that $P(k+1)$ is true whenever $P(k)$ is true.
- (d) If $P(k+1)$ is true then $P(n)$ is true for all values of n .

LOGARITHM

- **Logarithm**: Logarithm is the opposite of powers.
If $a = b^c$ then its logarithmic form is $\log_b a = c$ (log of a to the base b is c).
It is defined only for $a > 0$.
- **Common Logarithm**: Logarithm with base 10 is called as common logarithm. It is denoted by $\log$.
- **Natural Logarithm**: Logarithm with base e is called as natural logarithm. It is denoted by $\ln$.
- $\log_a(mn) = \log_a(m) + \log_a(n)$.
- $\log_a(m/n) = \log_a(m) - \log_a(n)$.
- $\log_a(m^n) = n\log_a(m)$.
- $\log_a(1) = 0$
- $\log_a(1/x) = -\log_a(x) = \log_{\frac{1}{a}}(x)$.
- $x = e^{\ln x}$.
- $\log_a(x) = \frac{\log_b(x)}{\log_b(a)}$.
- $\log_a(x) = \frac{1}{\log_x(a)}$.
- $\log_a(a) = 1$.
- $a^{\log_a(x)} = x$.
- $\log 10 = 1$.
- $\log 10^n = n$.
- $y^{\log_a(x)} = x^{\log_a(y)}$.
- $\log_{a^n}(x) = \frac{\log_a x}{n}$.
- $\log_{a^n}(x^n) = \log_a(x)$.

PERMUTATION & COMBINATION

- **Factorial Notation:** The product of first n natural numbers is called as factorial n and is denoted by $n! = \underline{n} = 1 \times 2 \times 3 \times 4 \times \dots \times n$.
 - (1) We have to assume $0! = 1$, though it makes no sense.
 - (2) When n is a negative integer then $n! = \underline{n}$ is not defined.
 - (3) $n! = \underline{n} = 1 \times 2 \times 3 \times 4 \times \dots \times n = n \times 1 \times 2 \times 3 \times 4 \times \dots \times (n-1)$
 $\Rightarrow n! = n \times (n-1)!$
 - (4) $(2n)! = 2^n \times n! \times (1 \times 3 \times 5 \times \dots \times (2n-1))$.
- **Multiplication Principle:** If an operation can be performed in ' m ' different ways and second operation can be performed in ' n ' different ways, then the two operations in succession can be performed in $m \times n$ different ways.
- **Addition Principle:** If an operation can be performed in ' m ' different ways and second operation which is independent of first operation, can be performed in ' n ' different ways, then the two operations can be performed in $m + n$ different ways.
- **Permutation:** Permutation means arrangement. Each of the different arrangements which can be made by taking some or all of given number of things or objects at a time is called a permutation.
- **Combination:** It means selection, Chose, or making committee. Each of the different groups or selections which can be made by taking some or all of a number of things (irrespective of order) is called as combination.
- ${}^n P_r = P(n, r) = \frac{n!}{(n-r)!} = n(n-1)(n-2)(n-3)\dots(n-r+1), 0 \leq r \leq n$.
- ${}^n P_r$ is always a positive integer.
- $\frac{{}^n P_r}{{}^n P_{-1r}} = n - r + 1$.
- ${}^n P_r = n \cdot {}^{n-1} P_{r-1} = n(n-1) {}^{n-2} P_{r-2} = n(n-1)(n-2) {}^{n-3} P_{r-3} = \dots$
- ${}^n P_r + r {}^n P_{r-1} = {}^{n+1} P_r$.
- ${}^n P_r = {}^{n-1} P_r + r {}^{n-1} P_{r-1}$.
- Number of permutations of ' n ' different things taken all at a time is ${}^n P_n = n!$.
- The number of permutations of ' n ' things taken all at a time, out of which ' p ' are alike and are of one type and ' q ' are alike and are of second type and rest all are different is $\frac{n!}{p! \cdot q!}$.
- The number of permutation of ' n ' different things taken ' r ' at a time when each thing may be repeated any number of times is n^r .
- The number of permutation of ' n ' different things taken ' r ' at a time, when a particular things is to be always included in each arrangement, is $r \cdot {}^{n-1} P_{r-1}$.

- The number of permutations of ' n ' different things taken ' r ' at a time, when ' s ' particular things are to be always included in each arrangement is ${}^r P_s \cdot {}^{n-1} P_{r-s} = r! \times {}^{n-s} C_{r-s}$.
- The number of permutation of ' n ' different things taken ' r ' at a time, when a particular things is never taken in each time is ${}^{n-1} P_r$.
- The number of permutation of ' n ' different things taken all at a time, when ' m ' specified things always come together is $m! \times (n-m+1)!$.
- The number of permutation of ' n ' different things taken all at a time, when ' m ' specified things never come together is $n! - (m! \times (n-m+1)!)$.
- The number of permutation of ' n ' different things taken ' r ' at a time, when atleast one thing is repeated is $n^r - {}^n P_r$.
- The number of permutation of ' n ' different things taken ' r ' at a time, when each thing may occur any number of times is $\frac{n(n^r - 1)}{n-1}$.
- Let there are ' n ' objects, of which ' m ' objects are alike of one kind, and the remaining ' $n-m$ ' objects are alike of another kind. Then the total number of distinct permutations that can be formed from these objects is $\frac{n!}{m! \times (n-m)!}$.
- Let there are ' n ' objects, of which ' p_1 ' objects are alike of one kind, ' p_2 ' are alike of 2nd kind, ..., ' p_k ' are alike of k th kind such that $p_1 + p_2 + \dots + p_k = n$. Then the total number of distinct permutations that can be formed from these objects is $\frac{n!}{(p_1!) \times (p_2!) \times \dots \times (p_k!)}$.
- The number of significant numbers consisting of ' r ' digits and formed out of ' n ' digits including zero (0), no digit being repeated in any number is ${}^n P_r - {}^{n-1} P_{r-1}$.
- The number of permutations of ' n ' different things where order of ' r ' things is not to be considered is $\frac{n!}{r!}$.
- The number of circular permutations of ' n ' different things is $(n-1)!$.
- The number of circular arrangements of ' n ' different things when clock wise and anti-clock wise arrangements are not different is $\frac{(n-1)!}{2}$.
- The number of circular arrangements of ' n ' different things taken when ' r ' at a time, when clock wise and anti-clock orders are taken as different is $\frac{{}^n P_r}{r}$.
- The number of circular arrangements of ' n ' different things taken when ' r ' at a time, when clock wise and anti-clock orders are not different is $\frac{{}^n P_r}{2r}$.
- **Derangement (or) Matching Problem:** If ' n ' things form an arrangement in a row, the number of ways in which they can be arranged so that no one of them occupies its original place is $n! \times \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \dots + (-1)^n \frac{1}{n!} \right)$.

- The sum of digits in unit places of all numbers formed with the help of $a_1, a_2, \dots, a_n$ all at a time is $(n-1)! \times (a_1 + a_2 + \dots + a_n)$.
- The sum of all possible numbers formed out of all the 'n' digits without zero is $= (n-1)! \times (\text{sum of all the digits}) \times (111\dots n \text{ times})$.

$$= (n-1)! \times (\text{sum of all the digits}) \times \left(\frac{10^n - 1}{9} \right)$$
- The sum of all possible numbers formed out of all the 'n' digits which includes zero is $= (\text{sum of all the digits}) [(n-1)! \times (111\dots n \text{ times}) - (n-2)! \times (111\dots (n-1) \text{ times})]$.

$$= (\text{sum of all the digits}) \left[(n-1)! \times \left(\frac{10^n - 1}{9} \right) - (n-2)! \times \left(\frac{10^{n-1} - 1}{9} \right) \right]$$
- Sum of all 'r' digits number formed by taking the given 'n' digits (including 0) is $= (\text{sum of all the digits}) [{}^{n-1}P_{r-1} \times (111\dots r \text{ times}) - {}^{n-2}P_{r-2} \times (111\dots (r-1) \text{ times})]$.
- Sum of all 'r' digits number formed by taking the given 'n' digits (excluding 0) is $= (\text{sum of all the digits}) \times {}^{n-1}P_{r-1} \times (111\dots r \text{ times})$.
- The number of mappings (functions) that can be defined from a set A containing 'm' elements to set B containing 'n' elements is n^m .
- The number of one-one (injective) mappings (functions) that can be defined from a set A containing 'm' elements to set B containing 'n' elements is nP_m where $m \leq n$.
- The number of one-one onto (bijective) mappings (functions) that can be defined from a set A containing 'n' elements to set B containing 'n' elements is ${}^nP_n = n!$.
- The number of onto (surjective) mappings (functions) that can be defined from a set A containing 'm' elements to set B containing 'n' elements or the number of 'n' different things can be distributed into 'r' different groups where each group must have atleast one thing is ${}^nC_1(n-1)^m + {}^nC_2(n-2)^m + {}^nC_3(n-3)^m + \dots$ where ${}^nC_r = 0$ for $n < r$.
- ${}^nC_r = \binom{n}{r} = C(n, r) = \frac{n!}{(n-r)!r!} = \frac{{}^nP_r}{r!}$.
- ${}^nC_r = \frac{n(n-1)(n-2)\dots(n-r+1)}{r(r-1)(r-2)\dots 1}$.
- ${}^nC_0 = {}^nC_n = 1$.
- ${}^nC_{n-1} + {}^nC_1 = n$.
- ${}^nC_{r-1} + {}^nC_r = {}^{n+1}C_r$.
- ${}^nC_r = {}^nC_{n-r}$.
- $\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}$.
- $\frac{{}^nC_r}{{}^{n-}C_{r-1}} = \frac{n}{r}$.
- $\frac{{}^nC_{r-1}}{{}^{n-1}C_{r-1}} = \frac{n}{n-r+1}$.

- If ${}^nC_r = {}^nC_s$ then either $r = s$ or $r + s = n$.
- If ${}^nC_{r-1}, {}^nC_r, {}^nC_{r+1}$ are in A.P. then $(n - 2r)^2 = (n + 2)$.
- If ' n ' is even, then greatest value of nC_r is ${}^nC_{n/2}$.
- If ' n ' is odd, then greatest value of nC_r is ${}^nC_{(n-1)/2}$ or ${}^nC_{(n+1)/2}$.
- ${}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n = 2^n$.
- ${}^nC_0 + {}^nC_2 + {}^nC_4 + \dots = {}^nC_1 + {}^nC_3 + {}^nC_5 + \dots = 2^{n-1}$.
- The number of combination of ' n ' different things taken ' r ' at a time, when any object may be repeated any number of times:

$$= \text{Coefficient of } x^r \text{ in } (1 + x + x^2 + x^3 + \dots + x^r)^n = \text{Coefficient of } x^r \text{ in } (1 - x)^{-n} = {}^{n+r-1}C_r$$
- The total number of ways in which it is possible to make groups by taking some or all out of $n = n_1 + n_2 + n_3 + \dots + n_r$ things when n_1 are alike of one kind, n_2 are alike of 2nd kind and n_r are alike of r th kind is $(n_1 + 1)(n_2 + 1)(n_3 + 1) \dots (n_r + 1) - 1$.
- The number of combination of ' n ' different things taken ' r ' at a time, when ' p ' particular things is to be always included, is ${}^{n-p}C_{r-p}$.
- The number of combination of ' n ' different things taken ' r ' at a time, when ' p ' particular things are never included, is ${}^{n-p}C_r$.
- The number of combination of ' n ' different things taken ' r ' at a time, when ' p ' particular things are not together in any selection, is ${}^nC_r - {}^{n-p}C_{r-p}$.
- The number of selections of ' r ' consecutive things out of ' n ' things in a row is $= n - r + 1$.
- The number of selections of ' r ' consecutive things out of ' n ' things along a circle is $= n$ if $(r < n)$.
- The number of selections of ' r ' consecutive things out of ' n ' things along a circle is $= 1$ if $(r = n)$.
- The number of combinations of ' n ' different things taken any number (0 or more) at a time is $= 2^n$ i.e. ${}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n = 2^n$.
- The number of combinations of ' n ' different things taken 1 or more at a time is $= {}^nC_1 + {}^nC_2 + \dots + {}^nC_n = 2^n - 1$.
- The number of selection of ' r ' things out of ' n ' identical things is $= 1$.
- The number of selection of 0 or more things out of ' n ' identical things is $= n + 1$.
- The number of selection of 1 or more things out of ' n ' identical things is $= n$.
- The number of combinations of ' n ' objects of which ' p ' are identical, taken ' r ' at a time is
$$= \begin{cases} {}^{n-p}C_r + {}^{n-p}C_{r-1} + {}^{n-p}C_{r-2} + \dots + {}^{n-p}C_0 & r \leq p \\ {}^{n-p}C_r + {}^{n-p}C_{r-1} + {}^{n-p}C_{r-2} + \dots + {}^{n-p}C_{r-p} & r > p \end{cases}$$
- The number of ways in which ' n ' different things can be arranged into ' r ' different groups is $= {}^{n+r-1}P_n = n! \times {}^{n-1}C_{r-1}$ according as blanks group are not admissible.
- The number of ways in which ' n ' different things can be distributed into ' r ' different groups is $= r^n - {}^rC_1(r-1)^n + {}^rC_2(r-2)^n - \dots + (-1)^{r-1} {}^rC_{r-1}$ according as blanks group are not admissible.

- The number distribution of ' n ' different things into ' r ' different groups in which no one of ' x ' of the ' x ' assigned parcels are blanks is $= r^n - {}^rC_1(r-1)^n + {}^rC_2(r-2)^n - \dots + (-1)^{r-1} {}^rC_{r-1}$.
- The number of ways in which $m+n$ different things can be divided equally into ' n ' groups is $= \frac{(m \times n)!n!}{(m!)^n} = \frac{(m \times n)!}{(m!)^n}$.
- The number of ways in which $m+n$ different things can be divided into two different groups of ' m ' and ' n ' things separately is $= {}^{m+n}C_m {}^nC_n = \frac{(m+n)!}{m!.n!}$ where $m \neq n$.
- The number of ways in which $m+n+p$ different things can be divided into three different groups of ' m ', ' n ' and ' p ' things separately is $= \frac{(m+n+p)!}{m!.n!.p!}$ where $m \neq n \neq p$.
- The number of ways of dividing ' $2n$ ' different things into two groups each containing ' n ' things and order of the groups is not important is $= \frac{(2n)!}{2!.(n!)^2}$.
- The number of ways of dividing ' $2n$ ' different things into two groups each containing ' n ' things and order of the groups is important is $= \frac{(2n)!}{(n!)^2}$.
- The number of ways of dividing ' $3n$ ' different things into three groups each containing ' n ' things and order of the groups is not important is $= \frac{(3n)!}{3!.(n!)^3}$.
- The number of ways of dividing ' $3n$ ' different things into three groups each containing ' n ' things and order of the groups is important is $= \frac{(3n)!}{(n!)^3}$.
- The number of ways of dividing ' kn ' different things into ' k ' groups each containing ' n ' things and order of the groups is not important is $= \frac{(kn)!}{k!.(n!)^k}$.
- The number of ways of dividing ' kn ' different things into ' k ' groups each containing ' n ' things and order of the groups is important is $= \frac{(kn)!k!}{(n!)^k k!} = \frac{(kn)!}{(n!)^k}$.
- If out of $p+q+r+t$ things ' p ' are alike of one kind, ' q ' are alike of second kind, ' r ' are alike of third kind and ' t ' are different, then the total number of selection is $= (p+1)(q+1)(r+1)2^t - 1$.
- If out of $p+q+r$ things ' p ' are alike of one kind, ' q ' are alike of second kind, ' r ' are alike of third kind, then the total number of selection is $= (p+1)(q+1)(r+1) - 1$.
- Let $n = a_1^p . a_2^q . a_3^r \dots \neq 1$ where $a_1, a_2, a_3, \dots$ are primes and $p, q, r \in N$, then number of all positive integers which are less than ' n ' and are prime to it is $= N \left(1 - \frac{1}{a_1}\right) \left(1 - \frac{1}{a_2}\right) \left(1 - \frac{1}{a_3}\right) \dots$.
- Let $n = a_1^p . a_2^q . a_3^r \dots \neq 1$ where $a_1, a_2, a_3, \dots$ are primes and $p, q, r \in N$, then number of divisors of ' n ' is $= (p+1)(q+1)(r+1) \dots$.

- Let $n = a_1^p . a_2^q . a_3^r \dots \neq 1$ where $a_1, a_2, a_3 \dots$ are primes and $p, q, r \in N$, then number of divisors of ' n ' excluding '1' is $= (p+1)(q+1)(r+1) \dots - 1$.
- Let $n = a_1^p . a_2^q . a_3^r \dots \neq 1$ where $a_1, a_2, a_3 \dots$ are primes and $p, q, r \in N$, then number of divisors of ' n ' excluding ' n ' is $= (p+1)(q+1)(r+1) \dots - 1$.
- Let $n = a_1^p . a_2^q . a_3^r \dots \neq 1$ where $a_1, a_2, a_3 \dots$ are primes and $p, q, r \in N$, then number of proper divisors (non trivial divisors) of ' n ' is $= (p+1)(q+1)(r+1) \dots - 2$.
- Let $n = a_1^p . a_2^q . a_3^r \dots \neq 1$ where $a_1, a_2, a_3 \dots$ are primes and $p, q, r \in N$, then number of improper divisors (trivial divisors) of ' n ' is $= 2$.
- Let $n = a_1^{p_1} . a_2^{p_2} . a_3^{p_3} \dots a_k^{p_k} \neq 1$ is a positive integer where $a_1, a_2, a_3 \dots, a_k$ are primes and $p_1, p_2, p_3 \dots, p_k \in N$ then the number of divisors of ' n ' is $= (p_1+1)(p_2+1)(p_3+1) \dots (p_k+1)$.
- Let $n = a_1^{p_1} . a_2^{p_2} . a_3^{p_3} \dots a_k^{p_k} \neq 1$ is a positive integer where $a_1, a_2, a_3 \dots, a_k$ are primes and $p_1, p_2, p_3 \dots, p_k \in N$ then the sum of divisors of ' n ' is $= \frac{a_1^{(p_1+1)} - 1}{a_1 - 1} \times \frac{a_2^{(p_2+1)} - 1}{a_2 - 1} \times \frac{a_3^{(p_3+1)} - 1}{a_3 - 1} \times \dots \times \frac{a_k^{(p_k+1)} - 1}{a_k - 1}$.
- Let $n = a_1^{p_1} . a_2^{p_2} . a_3^{p_3} \dots a_k^{p_k}$ where $a_1, a_2, a_3 \dots, a_k$ are primes other than 2 then number of even divisors is $= p_1(p_2+1)(p_3+1) \dots (p_k+1)$.
- Let $n = a_1^{p_1} . a_2^{p_2} . a_3^{p_3} \dots a_k^{p_k}$ where $a_1, a_2, a_3 \dots, a_k$ are primes other than 2 then number of odd divisors is $= (p_2+1)(p_3+1) \dots (p_k+1)$.
- Let $n = a_1^{p_1} . a_2^{p_2} . a_3^{p_3} \dots a_k^{p_k}$ where $a_1, a_2, a_3 \dots, a_k$ are primes other than 2 then number of ways in which ' n ' can be resolved into 2 factors if ' n ' is not a perfect square is $= \frac{1}{2}(p_1+1)(p_2+1)(p_3+1) \dots (p_k+1)$.
- Let $n = a_1^{p_1} . a_2^{p_2} . a_3^{p_3} \dots a_k^{p_k}$ where $a_1, a_2, a_3 \dots, a_k$ are primes other than 2 then number of ways in which ' n ' can be resolved into 2 factors if ' n ' is a perfect square is $= \frac{1}{2}\{(p_1+1)(p_2+1)(p_3+1) \dots (p_k+1) + 1\}$.
- The number of ways in which a composite number N can be resolved into two factors which are relatively prime (or co-prime) to each other is equal to 2^{n-1} where n is the number of different factors in N .
- The number of lines formed out of ' n ' points in a plane of which no three points are collinear is $= {}^nC_2$.
- The number of lines formed out of ' n ' points in a plane of which ' m ' points are collinear is $= {}^nC_2 - {}^mC_2 + 1$.
- The number of triangles formed out of ' n ' points in a plane of which ' m ' points are collinear is $= {}^nC_3 - {}^mC_3$.
- The number of diagonals of a polygon of ' n ' sides is $= {}^nC_2 - n = \frac{n(n-3)}{2}$.
- The number of parallelograms that can be formed from a set of ' n ' parallel lines intersecting another set of ' m ' parallel lines is $= {}^nC_2 \times {}^mC_2$.

- If 'n' distinct points are given on the circumference of a circle, then:
 - (1) No. of Straight lines $= {}^nC_2$.
 - (2) No. of Trinagles $= {}^nC_3$.
 - (3) No. of Quadrilaterals $= {}^nC_4$.
- The number of rectangles of any size in a square of $n \times n$ is $\sum_{r=1}^n r^3$.
- The number of squares of any size in a square of $n \times n$ is $\sum_{r=1}^n r^2$.
- The number of rectangles of any size of $n \times p$ where $n < p$ is $= {}^nC_2 \times {}^pC_2 = \frac{np(n+1)(p+1)}{4}$.
- The number of squares of any size of $n \times p$ where $n < p$ is $\sum_{r=1}^n (n+1-r)(p+1-r)$.
- The number of ways in which 'n' identical things can be distributed into 'r' different groups is ${}^{n+r-1}C_{r-1}$.
- The number of ways in which 'n' identical things can be distributed into 'r' different groups so that each group to receive at-least one things is ${}^{n-1}C_{r-1}$.
- The number of non-negative integral solutions of $x_1 + x_2 + x_3 + \dots + x_r = n$ is ${}^{n+r-1}C_{r-1}$.
- The number of positive integral solutions of $x_1 + x_2 + x_3 + \dots + x_r = n$ is ${}^{n-1}C_{r-1}$.
- The number of ways of selection of 'r' things from 'n' different groups when each group may have identical things of 'r' or more things is ${}^{n+r-1}C_r$.
- **Exponent of Prime p in n!** : Exponent of p in n! is denoted by

$E_p(n!) = \left[\frac{n}{p} \right] + \left[\frac{n}{p^2} \right] + \left[\frac{n}{p^3} \right] + \dots + \left[\frac{n}{p^k} \right]$ where $p^k < n < p^{k+1}$ and $\left[\frac{n}{p} \right]$ denotes the greatest integer function.

Example: Exponent of 3 in $100! = E_3(100!) = \left[\frac{100}{3} \right] + \left[\frac{100}{3^2} \right] + \left[\frac{100}{3^3} \right] + \left[\frac{100}{3^4} \right]$
 $\Rightarrow E_3(100!) = \left[\frac{100}{3} \right] + \left[\frac{100}{9} \right] + \left[\frac{100}{27} \right] + \left[\frac{100}{81} \right]$
 $\Rightarrow E_3(100!) = 33 + 11 + 3 + 1 = 48$ (ANS)

- $(n!)!$ is divisible by $(n!)^{(n-1)!}$.
- $(kn)!$ is divisible by $(n!)^k$.
- nC_r is divisible by n if n is a prime and $n \neq r$.
- **Multinomial Theorem:** Let $x_1, x_2, x_3, \dots, x_m$ be integers. Then number of solution to the equation $x_1 + x_2 + x_3 + \dots + x_m = n$ subject to the condition $a_1 \leq x_1 \leq b_1, a_2 \leq x_2 \leq b_2, \dots, a_m \leq x_m \leq b_m$ is equal to the coefficient of x^n in the expansion of: $(x^{a_1} + x^{a_1+1} + \dots + x^{b_1})(x^{a_2} + x^{a_2+1} + \dots + x^{b_2}) \dots (x^{a_m} + x^{a_m+1} + \dots + x^{b_m})$.

- The number of integral solution of $x_1 + x_2 + x_3 + \dots + x_m = n$ where $x_1 \geq 0, x_2 \geq 0, \dots, x_m \geq 0$ is same as the number of ways to distribute n identical things among m persons
 - = the coefficient of x^n in the expansion of: $(1 + x + x^2 + x^3 + \dots)^m$
 - = coefficient of x^n in $\left(\frac{1}{1-x}\right)^m$
 - = coefficient of x^n in $(1-x)^{-m}$
 - = coefficient of x^n in $(1 + {}^m C_1 x + {}^{m+1} C_2 x^2 + {}^{m+2} C_3 x^3 + \dots)$
 - = ${}^{m+n-1} C_n$
 - = ${}^{m+n-1} C_{m-1}$.
- The number of natural solution of $x_1 + x_2 + x_3 + \dots + x_m = n$
 - = the coefficient of x^n in the expansion of: $(x + x^2 + x^3 + \dots)^m$
 - = the coefficient of x^n in the expansion of: $x^m (1 + x + x^2 + x^3 + \dots)^m$
 - = the coefficient of x^{n-m} in the expansion of: $(1 + x + x^2 + x^3 + \dots)^m$
 - = coefficient of x^{n-m} in $\left(\frac{1}{1-x}\right)^m$
 - = coefficient of x^{n-m} in $(1-x)^{-m}$
 - = coefficient of x^{n-m} in $(1 + {}^m C_1 x + {}^{m+1} C_2 x^2 + {}^{m+2} C_3 x^3 + \dots)$
 - = ${}^{m+n-m-1} C_{n-m}$
 - = ${}^{n-1} C_{m-1}$
- The number of possible arrangement of n objects out of which ' p_1 ' objects are alike of one kind, ' p_2 ' are alike of 2nd kind, ..., ' p_k ' are alike of k th kind such that $p_1 + p_2 + \dots + p_k = n$ is:
 - = $(n!) \times$ Coefficient of x^n in the expansion of $\times$

$$\left(1 + x + \frac{x^2}{2!} + \dots + \frac{x^{p_1}}{p_1!}\right) \left(1 + x + \frac{x^2}{2!} + \dots + \frac{x^{p_2}}{p_2!}\right) \dots \left(1 + x + \frac{x^2}{2!} + \dots + \frac{x^{p_k}}{p_k!}\right).$$

• **The Pigeon-Hole Principle:**

- (1) If more than n objects are distributed into n compartments, then atleast one compartment must receive more than one objects.
- (2) If $r.n + 1$ ($r \geq 1$) pigeons are distributed among n pigeon holes, one of the pigeon holes will contain atleast $r + 1$ pigeons.
- (3) If n pigeons are placed in m pigeon holes, then atleast one pigeon hole will contain more than $\left\lceil \frac{n-1}{m} \right\rceil$ pigeons, where $[x]$ denotes the greatest integer x .

